PML-AI
Body of Knowledge

The reference for the PCI Project Management Leader – AI
Leadership · delivery systems · governance · the governed use of AI

FIRST EDITION — PHASE 1 PROTOTYPE (NOT FOR RELEASE)
PROJECT CONTROLS INSTITUTE GLOBAL

Finance intelligently. Control predictively. Deliver successfully.

AI proposes; the professional verifies, decides and remains accountable.

Prototype notice

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PART TWO

Delivering the work

Domains 5–10 — scope, planning and scheduling, cost and commercial awareness, risk, quality and procurement: the delivery disciplines a leader integrates.

06

DOMAIN 6

Planning, Scheduling and Delivery Flow (quantitative flagship)

Group: Delivering the work (Domain 6 of 6 in Part Two). **Target:** ~76 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is the home of the schedule symbols — `ES`, `EF`, `LS`, `LF`, `TF` (total float), `FF` (free float) — restated by every domain that touches time. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

A project leader's authority rests on one repeated act: making a credible statement about the future — when the work will finish, what could move that date, and what it would cost to move it back. This domain builds the machinery behind that credibility. It starts where planning actually starts — with levels of plan and the logic that binds work together (KA 6.1); builds the critical path method in full, forward and backward passes, and both kinds of float (KA 6.2); adds the realities the textbook network ignores — resources, constraints, milestones and rolling-wave elaboration (KA 6.3); and finishes with delivery flow across predictive, agile and hybrid worlds: recovery, compression economics, and forecasting under uncertainty (KA 6.4). Cost joins schedule in Domain 7 (earned value); risk quantification deepens in Domain 8. A leader who cannot read a network diagram is hostage to whoever can; this domain removes the hostage-taking.

Learning objectives. After this domain a candidate can: choose the right planning level for an audience and a decision; build a logic network with correct dependency types; run forward and backward passes and derive total and free float; identify the critical path and near-critical paths and explain why float is not spare time; plan resources against the schedule and level them deliberately; apply rolling-wave elaboration honestly; select and price schedule compression; distinguish predictive, agile and hybrid scheduling and govern each; and use three-point estimates and scenario analysis to forecast completion with stated uncertainty — with any AI-generated schedule subject to the family's verification rule.

The master worked project. One project runs through this domain and returns in Domains 7 and 8: **Project Auriga**, a control-systems upgrade for a regional utility, planned in weeks:

ID	Activity	Duration (wk)	Predecessors
A	Mobilise	2	—
B	Detailed design	6	A
C	Procure control hardware	8	B
D	Civil and cabling works	7	B
E	Installation	5	C, D
F	Testing and commissioning	4	E
G	Operator training and handover	2	D

Every calculation below uses this network.

KNOWLEDGE AREA 6.1

Planning levels and logic networks

Topics: 6.1.1 levels of plan · 6.1.2 dependencies and logic · 6.1.3 network rules and quality.

6.1.1 Levels of plan

The principle. One schedule cannot serve a board, a site supervisor and a lender at once. Mature delivery organisations maintain a **hierarchy of schedules**, each derived from the one below, each honest with the one above:

Level	Name	Audience & decision	Typical granularity
L1	Executive summary	Board, sponsor — go/no-go, milestones	10–30 bars
L2	Management schedule	Steering, PMO — stage gates, interfaces	50–300 activities
L3	Control schedule	Project team — the CPM network; progress and float live here	300–5,000 activities
L4	Execution/lookahead	Site and squads — 2–6 week detail	daily/shift detail

The **control schedule (L3)** is where this domain's machinery operates: it carries the logic, the floats and the critical path, and it is the version under change control (Domain 4, KA 4.3). An L1 milestone that cannot be traced to L3 logic is decoration, and steering committees should treat it as such (Domain 3, KA 3.2).

Fig 6.1.1 — The schedule hierarchy: one logic, four honest views

6.1.2 Dependencies and logic

Definitions. Activities link through four dependency types — finish-to-start (**FS**, the default: design finishes, then procurement starts), start-to-start (**SS**: cabling can start once civils have started, often with a lag), finish-to-finish (**FF**: commissioning documentation finishes when commissioning finishes), and the rare start-to-finish (**SF**). A **lag** delays a successor (`FS+2` = start two weeks after the predecessor finishes); a **lead** (negative lag) overlaps it. Logic expresses physics and choice — what must follow, and what the team has chosen to sequence — and the two should be distinguishable in the schedule's notes, because only chosen logic can be re-chosen when recovery demands it (KA 6.4).

Common pitfall — the lag that hides work. A 3-week lag labelled "curing" is physics; a 3-week lag labelled nothing is often an activity somebody didn't model — unresourced, uncosted, invisible to progress measurement. Audit rule: every lag has a stated reason, and long lags (> 10 % of project duration) are activities in disguise.

WORKED EXAMPLE

Worked example 6.1.2 — re-choosing a dependency.

1. **Setup.** Survey work P (4 weeks) precedes report drafting Q (6 weeks), currently linked `FS+2` (two weeks of data cleaning after all surveying ends). The team proposes drafting in parallel as survey sections complete: `SS+1`. Compute Q's dates both ways (P starts week 0).
2. **Formula.** FS+lag: `ES(Q) = EF(P) + lag`. SS+lag: `ES(Q) = ES(P) + lag`.
3. **Substitution.** FS+2: `ES(Q) = 4 + 2 = 6`, `EF(Q) = 12`. SS+1: `ES(Q) = 0 + 1 = 1`, `EF(Q) = 7`.
4. **Result.** The chain finishes week **12** under FS+2 and week **7** under SS+1 — five weeks earlier from one logic choice.
5. **Interpretation.** Nothing was crashed and nobody works faster: the saving comes from overlapping, and its price is rework risk if early sections change after later surveying. This is why chosen logic must be visibly chosen (not fossilised as physics) — it is the cheapest recovery currency the project owns (KA 6.4.2), and spending it is a risk decision, not a scheduling trick.

6.1.3 Network rules and quality

A network the passes can trust obeys checkable rules: every activity except the first has a predecessor and except the last a successor (**no dangles**); no loops; durations estimated by the owning discipline, not the scheduler; no date constraints doing logic's job (a "must start on" pin that overrides logic converts the schedule from a model into a poster); and logic density in a healthy range — too few links and float is fiction, too many and nothing can move. These rules are the schedule-quality gate a leader can run without scheduling software: ask for the dangle count, the constraint count and the reasons for both.

AI in this KA

Scheduling tools now draft networks from scope statements and historical libraries. Useful — the draft logic is often 80 % right — and dangerous in a specific way: generated logic looks authoritative while encoding assumptions nobody made. The governed workflow: AI proposes the network; the discipline leads walk every link of their scope ("does B truly need all of A finished?"); the scheduler runs the quality rules above; and the leader signs the logic as a decision record (Domain 3, KA 3.3.4). **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 6.1

Term	Meaning
Schedule hierarchy (L1–L4)	Nested plans for different audiences, derived from one logic.
Control schedule	The L3 CPM network under change control; home of float and progress.
FS / SS / FF / SF	The four dependency types.
Lag / lead	Imposed delay / overlap on a dependency.
Dangle	An activity missing a predecessor or successor; a network defect.
Date constraint	A pinned date overriding logic; each one weakens the model.

SAMPLE MCQS — KA 6.1

MCQ 6.1-A `[6.1.2 · Application]` Cabling may begin one week after civil works begin. The correct dependency is: - A. FS+1 - B. SS+1 - C. FF+1 - D. FS−1

Rationale: The condition binds the two starts with a one-week lag: start-to-start plus one. A would wait for all civils to finish; C binds the finishes; D (a lead on FS) overlaps the finish, which is not what was stated.

MCQ 6.1-B `[6.1.3 · Analysis]` A schedule shows 14 "must finish on" constraints, all on milestones reported to the board. The most accurate reading is: - A. the schedule is well-governed, because board dates are protected - B. the milestones will be met, because the tool will honour the constraints - C. the schedule may no longer model reality — pinned dates can mask logic-driven slippage that float analysis would otherwise reveal - D. constraints are neutral scheduling hygiene with no analytical effect

Rationale: Pins override logic: the network can be slipping while pinned milestones hold still, hiding negative float until it is unrecoverable. A and B mistake suppression for control; D is false — every pin degrades the model's predictive value.

MCQ 6.1-E `[6.1.1 · Recall]` The schedule level that carries the logic, the floats and the critical path — and sits under formal change control — is: - A. L1, because the board owns the schedule - B. L2, because the PMO maintains it - C. L3, the control schedule - D. L4, because it has the most detail

Rationale: The L3 control schedule is the analytical model; L1/L2 summarise it and L4 elaborates near-term execution from it. Detail (D) is not the same as control — L4 churns weekly by design and is never the baselined network.

MCQ 6.1-C `[6.1.2 · Application]` Survey P (4 wk) feeds report Q (6 wk). Under `FS+2` the chain completes week 12. Re-linked `SS+1`, it completes week: - A. 7 - B. 11 - C. 12 - D. 5

Rationale: `ES(Q) = ES(P) + 1 = 1`, `EF(Q) = 7`. B subtracts one lag week from 12; C assumes logic changes nothing; D forgets Q's own duration must still run.

MCQ 6.1-D `[6.1.3 · Recall]` A "dangle" in a schedule network is: - A. an activity with negative float - B. an activity missing a predecessor or successor link - C. a milestone with zero duration - D. a lag longer than its predecessor

Rationale: Dangles are unbound activity ends — the passes cannot constrain them, so their dates and float are unreliable. A describes constraint-driven lateness; C describes every milestone; D is unusual but legal when justified.

SELF-CHECK — KA 6.1

1. Why must chosen logic be distinguishable from physical logic? — Because recovery (KA 6.4) works by re-choosing choices; physics cannot be renegotiated.
2. What three numbers give a fast schedule-quality read? — Dangle count, constraint count, unexplained-lag count — each should be at or near zero, with reasons for the rest.

KNOWLEDGE AREA 6.2

The critical path and float

Topics: 6.2.1 the forward and backward passes · 6.2.2 total and free float · 6.2.3 reading the critical path.

6.2.1 The forward and backward passes

Definitions. For each activity: `ES`/`EF` — earliest start/finish (forward pass, left to right: `ES` = latest `EF` of predecessors; `EF = ES + duration`); `LS`/`LF` — latest start/finish that do not delay the project (backward pass, right to left: `LF` = earliest `LS` of successors; `LS = LF − duration`). The project duration is the largest `EF` in the network.

Worked example 6.2.1 — Auriga, both passes. The passes are inherently tabular; the labelled table replaces the five-step skeleton, and the interpretation follows (pattern rule).

Act	Dur	ES	EF	LS	LF	TF	FF
A	2	0	2	0	2	0	0
B	6	2	8	2	8	0	0
C	8	8	16	8	16	0	0
D	7	8	15	9	16	1	0
E	5	16	21	16	21	0	0
F	4	21	25	21	25	0	0
G	2	15	17	23	25	8	8

Interpretation. Project duration **25 weeks**; the **critical path is A–B–C–E–F** (2+6+8+5+4 = 25), every activity on it with zero float. D can slip one week without moving the end date; G can slip eight. The passes turn a drawing into an instrument: every schedule statement a leader makes in this domain — "we finish in week 25", "D matters more than G", "procurement drives the job" — is read directly off this table.

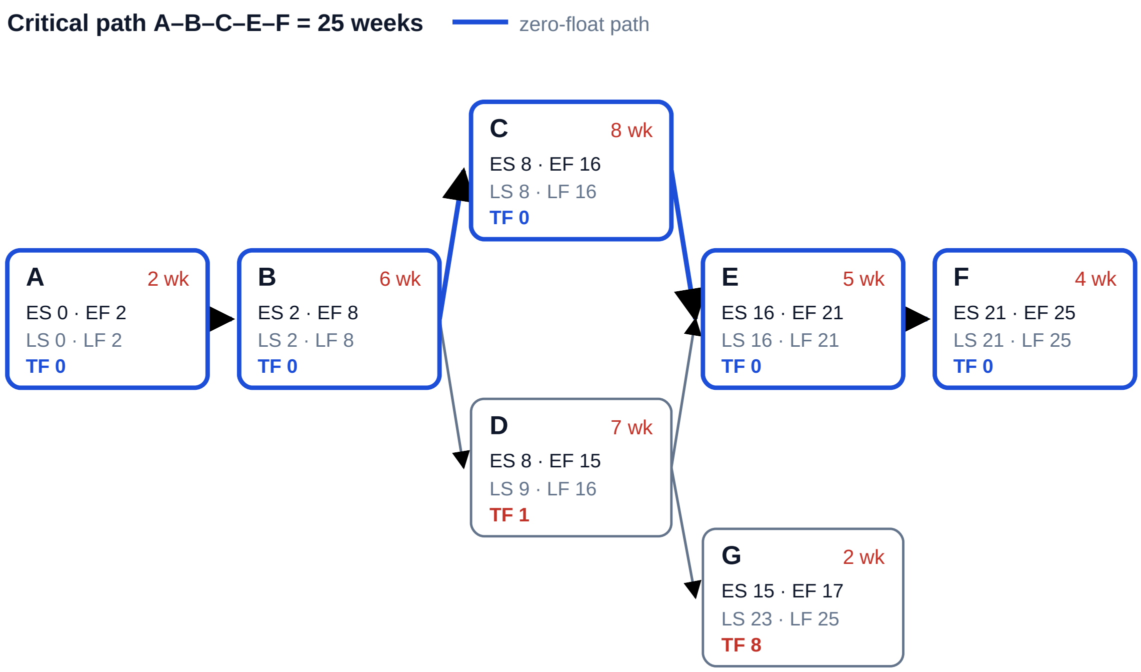

Fig 6.2.1 — The Auriga network with both passes

6.2.2 Total and free float

Definitions.

```
TF = LS − ES          — how far an activity can slip without delaying the project
FF = min(ES of successors) − EF   — how far it can slip without delaying ANY successor
```

`FF ≤ TF` always. The Auriga table carries the domain's subtlest lesson: **D has `TF` = 1 but `FF` = 0** — D can slip a week without moving the end date, but the moment it slips at all, G's earliest start moves. Total float is a project-level commodity; free float is a courtesy to your successors. A leader who grants a subcontractor "the float" without saying which kind has just given away someone else's schedule.

Float is not spare time. Float belongs to the path, not the activity: consume D's week and every activity sharing that path inherits zero. Contract regimes differ on who owns float (Domain 10, KA 10.3, treats float ownership clauses); the delivery answer is simpler — float is a risk buffer under the leader's governance, spent deliberately or not at all.

Common pitfall — managing only the critical path. D sits one week from critical. A team watching only A–B–C–E–F will discover D's importance the week it becomes too late to matter. Near-critical analysis (all paths within, say, 10 % of project duration) is standing practice; KA 6.4's scenario work builds on it.

6.2.3 Reading the critical path

The critical path is the longest path and therefore the shortest possible duration — both statements are the same fact, and fluency means holding both. Three leader-grade readings of the Auriga table: **(1)** procurement (C) is the single most schedule-driving activity — an expediting conversation is worth more than any amount of site pressure; **(2)** the C/D pair converge on E, so E's start is hostage to the later of two chains — convergence points are where slippage compounds and where progress review should focus (Domain 8, KA 8.1); **(3)** G's eight weeks of float make it the natural donor of resources in any recovery (KA 6.4).

AI in this KA

CPM arithmetic is deterministic — ideal for machine computation and instant human verification. The two checks that catch nearly every pass error: the critical path's durations must **sum exactly** to the project duration, and every critical activity must show `TF` = 0 with `ES = LS`. An AI-produced schedule that fails either is broken; one that passes both may still have wrong logic — which is why KA 6.1's link-walking discipline precedes trust in any pass, human or machine.

KEY TERMS — KA 6.2

Term	Meaning
Forward / backward pass	Left-to-right ES/EF computation; right-to-left LS/LF computation.
`ES EF LS LF`	Earliest/latest start and finish per activity.
Total float `TF`	`LS − ES`; slip available without delaying the project.
Free float `FF`	Slip available without delaying any successor; `FF ≤ TF`.
Critical path	The zero-float longest path; the project's shortest possible duration.
Convergence point	A node where paths merge; slippage compounds there.

SAMPLE MCQS — KA 6.2

MCQ 6.2-A `[6.2.1 · Application]` In the Auriga network, activity D (duration 7, predecessor B finishing week 8, successors E and G) has `LS` = 9 and `ES` = 8. Its total float is: - A. 0 - B. 1 - C. 8 - D. 7

Rationale: `TF = LS − ES = 9 − 8 = 1`. A confuses D with the critical activities; C is G's float; D is the activity's duration, not its float.

MCQ 6.2-B `[6.2.2 · Analysis]` An activity shows `TF` = 1 and `FF` = 0. Delaying it by one week will: - A. delay the project by one week - B. delay nothing, because total float absorbs the slip - C. delay at least one successor's earliest start while leaving the project end date unchanged - D. be impossible — free float can never be below total float

Rationale: Positive `TF` protects the end date (not A); zero `FF` means some successor moves immediately (not B). D inverts the invariant — `FF ≤ TF` always.

MCQ 6.2-C `[6.2.3 · Application]` Auriga's procurement activity C is crashed from 8 weeks to 6. The new project duration is: - A. 23 weeks - B. 24 weeks - C. 25 weeks - D. 21 weeks

Rationale: After one week of crashing, path B–D–E (8+7+5+4 via D) becomes co-critical at 24; the second crashed week buys nothing — duration stays 24. A assumes both weeks convert to project weeks; C forgets the crash entirely; D subtracts from the wrong baseline. (The full economics: KA 6.4.)

MCQ 6.2-F `[6.2.1 · Application]` Auriga's procurement C slips from 8 weeks to 9 (all else unchanged). The new project duration is: - A. 25 weeks - B. 26 weeks - C. 27 weeks - D. 24 weeks

Rationale: C was critical with zero float, so its extra week passes straight through: E runs 17–22, F finishes week 26. A assumes float absorbed a critical activity's slip; C adds the week twice; D subtracts it.

MCQ 6.2-D `[6.2.1 · Application]` Auriga's training activity G runs ES 15–EF 17 with `LF` = 25. Its total float is: - A. 8 - B. 10 - C. 2 - D. 0

Rationale: `TF = LF − EF = 25 − 17 = 8` (equivalently `LS − ES = 23 − 15`). B ignores G's duration (25 − 15); C is G's duration; D confuses G with the critical path.

MCQ 6.2-E `[6.2.3 · Analysis]` A programme's binding path shows `TF` = −3 after a constraint-honest pass. The professional reading is: - A. the software has malfunctioned — float cannot be negative - B. the plan, as constrained, is already three weeks late; the number sizes the recovery problem and must be reported now - C. delete the constraints so the float returns to zero - D. the project will finish three weeks early

Rationale: Negative float is the gap between what logic needs and what constraints allow — information, not error (A) and not good news (D). C is the pin-and-squeeze suppression this domain's Case B dissects; it hides the problem until it is unrecoverable.

SELF-CHECK — KA 6.2

1. State both definitions of the critical path and why they are equivalent. — Longest path through the network; shortest possible project duration — the longest chain of unavoidable work is precisely what bounds how soon the whole can finish.
2. Why is $FF \leq TF$ always? — Delaying any successor is one of the ways of delaying the project; the constraint set defining `FF` is stricter.
3. Auriga slips: D takes 10 weeks, not 7. New duration? — 27 weeks: D finishes week 18, E waits for it (18 > 16), E–F add 9 → the critical path re-routes through D (KA 6.4 works the recovery).

KNOWLEDGE AREA 6.3

Resources, constraints, milestones and rolling wave

Topics: 6.3.1 resource planning and levelling · 6.3.2 constraints and milestones · 6.3.3 rolling-wave planning.

6.3.1 Resource planning and levelling

The principle. The passes of KA 6.2 assume infinite resources; reality staffs the network. Loading resources onto the schedule produces a **histogram** — demand per period — and the histogram almost always spikes. Two responses: **resource smoothing** uses float to flatten demand without moving the end date (slip D and G inside their float to pull crews off the peak); **resource levelling** accepts a later end date if the resource cap is hard. The order of operations is a leadership decision in disguise: smoothing spends float — the project's risk buffer — to save money, and someone accountable should price that trade explicitly, not discover it in a progress meeting.

WORKED EXAMPLE

Worked example 6.3.1 — smoothing Auriga's field crews.

1. **Setup.** Weeks 9–15: civil works D needs 3 crews; early installation staging inside C needs 2; the site cap is 4 crews. Overlap weeks 9–15 demand 5 — one over cap.
2. **Formula.** Smoothing test: can a demanding activity slip within `TF` (and, if successors matter, `FF`) to clear the peak?
3. **Substitution.** D has `TF` = 1, `FF` = 0: slipping D one week clears one peak week but consumes D's entire float and immediately moves G (`FF` = 0). The staging work inside C has discretionary timing across weeks 9–16.
4. **Result.** Re-sequence the staging within C to weeks 14–16 where D's demand tails off: peak demand falls to 4, the cap holds, the end date holds, and **no float is spent**.
5. **Interpretation.** The cheapest capacity is re-sequencing; the second cheapest is float; the expensive ones are overtime, second shifts and extension. A leader asks for the histogram with the options priced, not just the spike.

WORKED EXAMPLE

Worked example 6.3.1b — when the cap is hard: levelling, priced.

1. **Setup.** Now suppose the site cap is **3 crews** (a permit condition, not a preference). D needs all 3 for weeks 9–15; the 4-week staging package (2 crews) must finish before E can start (week 16). Delay costs USD 45,000 per week (the client's bonus forgone); a second-shift waiver for the staging crew costs USD 20,000 per week. What does the leader do?
2. **Formula.** Compare total cost of each feasible plan: pure levelling (extend) vs paid capacity (second shift) vs any float-funded re-sequence.
3. **Substitution.** Levelling: staging cannot start until D releases crews (week 16); it runs weeks 16–19, E starts week 20 — project 25 → 29, cost `4 × 45,000 = 180,000`. Second shift: staging runs weeks 12–15 off-shift, E starts on time — cost `4 × 20,000 = 80,000`. No float-funded option exists: D's `TF` = 1 cannot host a 4-week move.
4. **Result. Buy the second shift: USD 80,000** against USD 180,000 of extension — and the end date holds.
5. **Interpretation.** A hard cap converts a scheduling problem into a procurement problem. The histogram's job is to surface that conversion early enough for the cheap option to exist — second shifts, pre-assembly, off-site staging all have lead times of their own. Levelling that silently extends the project is a decision someone should have been asked to make.

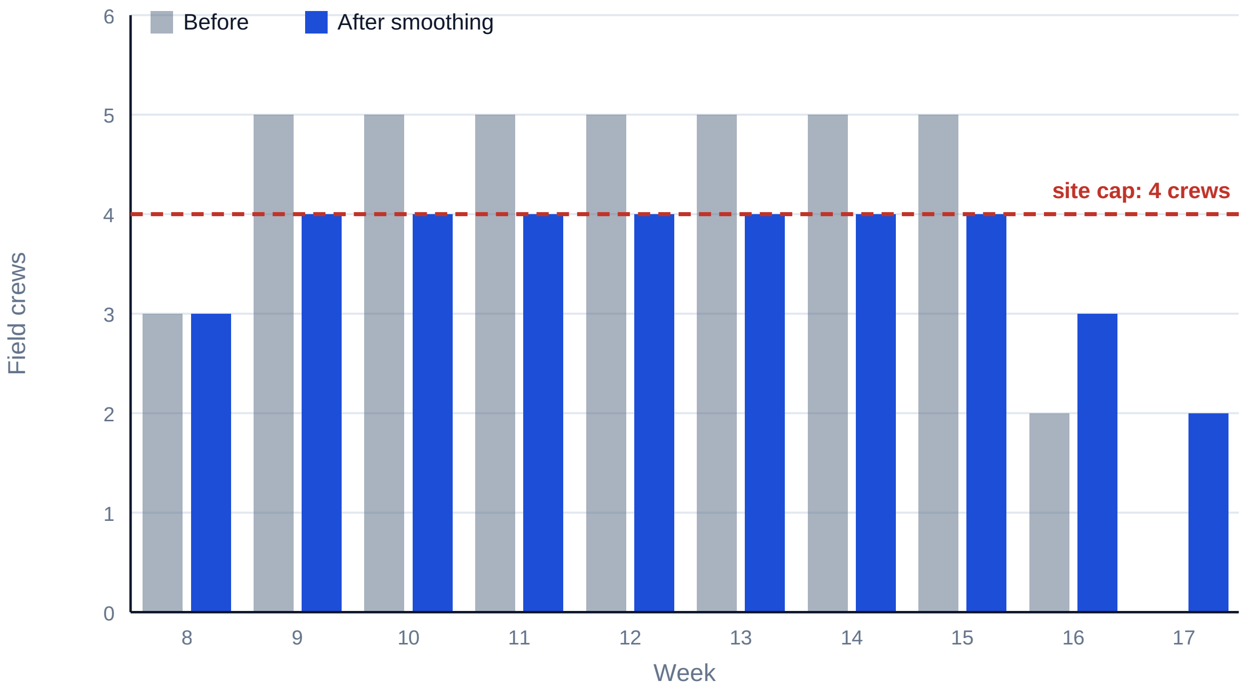

Fig 6.3.1 — Auriga field-crew histogram, before and after smoothing

6.3.2 Constraints and milestones

Constraints are externally imposed dates — an outage window, a regulatory deadline, a seasonal limit. Modelled honestly they are logic (an outage window is a predecessor with a date); modelled lazily they are pins that falsify float (KA 6.1.3). **Milestones** are zero-duration events that carry meaning: contractual obligations (Domain 10), gate decisions (Domain 3), integration points with other projects (Domain 15). A milestone's date is an output of the network, never an input — the moment a milestone is typed rather than computed, the schedule has stopped forecasting and started decorating.

Milestone hygiene: each has an owner, an unambiguous done-definition (Domain 5's acceptance criteria), traceable L3 logic, and — for contractual milestones — a stated float position reported honestly to the counterparty (Domain 11's reporting ethics).

6.3.3 Rolling-wave planning

The principle. Detail decays with distance: estimating month 18's tasks at daily granularity today manufactures precision, not knowledge. **Rolling wave** plans near-term work at execution detail (L4) and far work at planning packages (L3 summary), elaborating each wave as it approaches under change control.

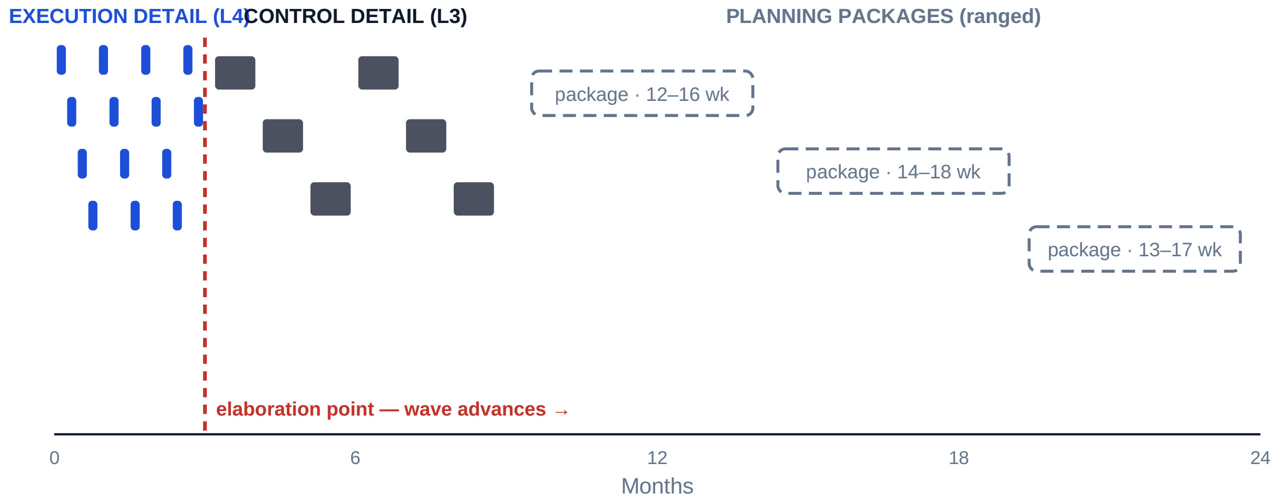

Fig 6.3.2 — The rolling-wave horizon

AI in this KA

Resource optimisation is a genuine AI strength — levelling across thousands of activities and dozens of calendars is combinatorial work machines do better than planners. Governance holds the same shape: the optimiser proposes a levelled plan; the planner verifies the constraint set was real (crew interchangeability, shift rules, site logistics); the leader owns the trade the optimiser cannot see — whether spending float, money or scope is the right currency this month. An optimiser given a wrong constraint produces a beautifully efficient fiction.

KEY TERMS — KA 6.3

Term	Meaning
Resource histogram	Demand per period from the loaded schedule.
Smoothing / levelling	Flattening demand within float / accepting a later date under a hard cap.
Constraint	An externally imposed date; modelled as logic, not as a pin.
Milestone	Zero-duration event with an owner and a done-definition; its date is computed.
Rolling wave	Near-term detail, far-term packages, elaborated under change control.
Planning package	A far-wave summary activity with ranged duration awaiting elaboration.

SAMPLE MCQS — KA 6.3

MCQ 6.3-A `[6.3.1 · Application]` A planner clears a resource peak by delaying an activity with `TF` = 1, `FF` = 0 by one week. The immediate consequences are: - A. no schedule effect of any kind - B. project delay of one week - C. the end date holds, the activity's path float is exhausted, and at least one successor's earliest start moves - D. the resource peak worsens

Rationale: One week inside `TF` protects the end date (not B), but zero `FF` moves a successor and the path's buffer is now spent (not A) — the smoothing was legal but not free.

MCQ 6.3-B `[6.3.3 · Analysis]` A 30-month programme shows daily-level tasks throughout, including month 30. The strongest inference is: - A. the planning team is unusually diligent - B. far-horizon detail is manufactured precision that will churn on every update, obscuring real variance - C. the schedule will need no re-baselining - D. rolling-wave planning has been correctly applied

Rationale: Detail beyond the knowable horizon creates update churn and false confidence — the opposite of diligence in effect (A) and the opposite of rolling wave (D); constant churn makes re-baselining more likely, not less (C).

MCQ 6.3-E `[6.3.1 · Application]` D needs 3 crews in weeks 9–15 and concurrent staging needs 2; the site cap is 4. The excess demand the histogram must clear is: - A. 1 crew for 7 weeks - B. 5 crews for 7 weeks - C. 2 crews for 7 weeks - D. nothing — 3 + 2 = 5 is within a 4-crew cap across two activities

Rationale: Peak demand 5 against cap 4 leaves one excess crew-week in each of the seven overlap weeks — the precise quantity smoothing must relocate. B is total demand, not excess; C is the staging demand itself; D misreads the cap as per-activity when it is per-site.

MCQ 6.3-C `[6.3.1 · Application]` A hard 3-crew cap forces either a 4-week project extension (delay cost USD 45,000/week) or a second-shift waiver at USD 20,000/week for the same 4 weeks (end date held). The value-maximising choice and its saving are: - A. extend: saves USD 100,000 - B. second shift: saves USD 100,000 - C. second shift: saves USD 25,000 - D. they cost the same

Rationale: Extension costs `4 × 45,000 = 180,000`; the shift premium `4 × 20,000 = 80,000` — choosing the shift saves USD 100,000 and keeps the date. A picks the dearer plan; C compares one week only; D ignores the 65,000-per-week difference... which compounds four times.

MCQ 6.3-D `[6.3.3 · Recall]` A legitimate planning package in a rolling-wave schedule must carry: - A. daily-level task detail for its whole span - B. a ranged duration consistent with its estimate class and a dated elaboration event under change control - C. zero float, to keep pressure on the team - D. a pinned finish date agreed with the sponsor

Rationale: Far-wave honesty is ranged duration plus a governed elaboration point. A is the manufactured precision rolling wave exists to avoid; C fabricates criticality; D is the typed milestone anti-pattern (Case study B).

SELF-CHECK — KA 6.3

1. Why is smoothing "spending the risk buffer"? — It consumes float, and float is the project's absorption capacity for the unknown (KA 6.2.2).
2. What makes a milestone honest? — Computed date, named owner, unambiguous done-definition, traceable logic.
3. What must a planning package carry to be legitimate? — A ranged duration consistent with its estimate class and a dated elaboration event under change control.

KNOWLEDGE AREA 6.4

Delivery flow: predictive, agile and hybrid; recovery and forecasting

Topics: 6.4.1 flow across delivery modes · 6.4.2 schedule compression and recovery · 6.4.3 forecasting and scenario analysis.

6.4.1 Flow across delivery modes

Predictive scheduling (this domain so far) plans the whole network and measures variance against baseline. **Agile** delivery (Domain 13 in full) fixes cadence and capacity and lets scope flow: the schedule questions become throughput (items per iteration), cycle time and forecast by velocity. **Hybrid** programmes — the commonest reality — run engineered streams predictively and product streams by cadence, joined at **integration milestones** that appear in the CPM network as fixed-capacity deliveries. The leader's job is translation: a velocity forecast ("the reporting module completes in 4–6 sprints") enters the network as a ranged duration on the integration activity, and the network returns what the programme needs from the agile stream — the latest acceptable delivery, read from `LS`. Neither mode is senior; the network prices time, the cadence protects focus, and governance (Domain 3, KA 3.1.3) holds both to evidence.

6.4.2 Schedule compression and recovery

The two levers. Crashing buys duration with money (more resources, overtime, expediting); **fast-tracking** buys it with risk (overlapping activities that logic preferred in sequence). Both obey the same law: work the critical path, and re-run the passes after every move, because **the critical path migrates**.

WORKED EXAMPLE

Worked example 6.4.2 — the crash that stopped paying.

1. **Setup.** Auriga's client offers a bonus of USD 45,000 (≈ SAR 168,750) per week saved. Expediting C costs USD 30,000 per week, up to two weeks. Should the leader buy one week, two, or none?
2. **Formula.** Crash only while (weeks actually saved × value per week) > crash cost — checked against the re-run passes, not the original network.
3. **Substitution.** Crash C by 1 (8→7): duration 25→24; net gain `45,000 − 30,000 = +15,000`. Crash C by 2 (8→6): path B–D–E–F (2+6+7+5+4) is now co-critical at 24 — duration stays 24; second week's net `0 − 30,000 = −30,000`.
4. **Result. Crash one week only**: +USD 15,000. The second week is pure loss.
5. **Interpretation.** Compression is subject to sharply diminishing returns because every crash promotes the next-longest path. The disciplined sequence: identify the binding path → price the cheapest week on it → re-run the passes → repeat until the next week costs more than it is worth. Leaders who mandate "crash procurement by two weeks" from the original network pay for weeks that no longer exist.

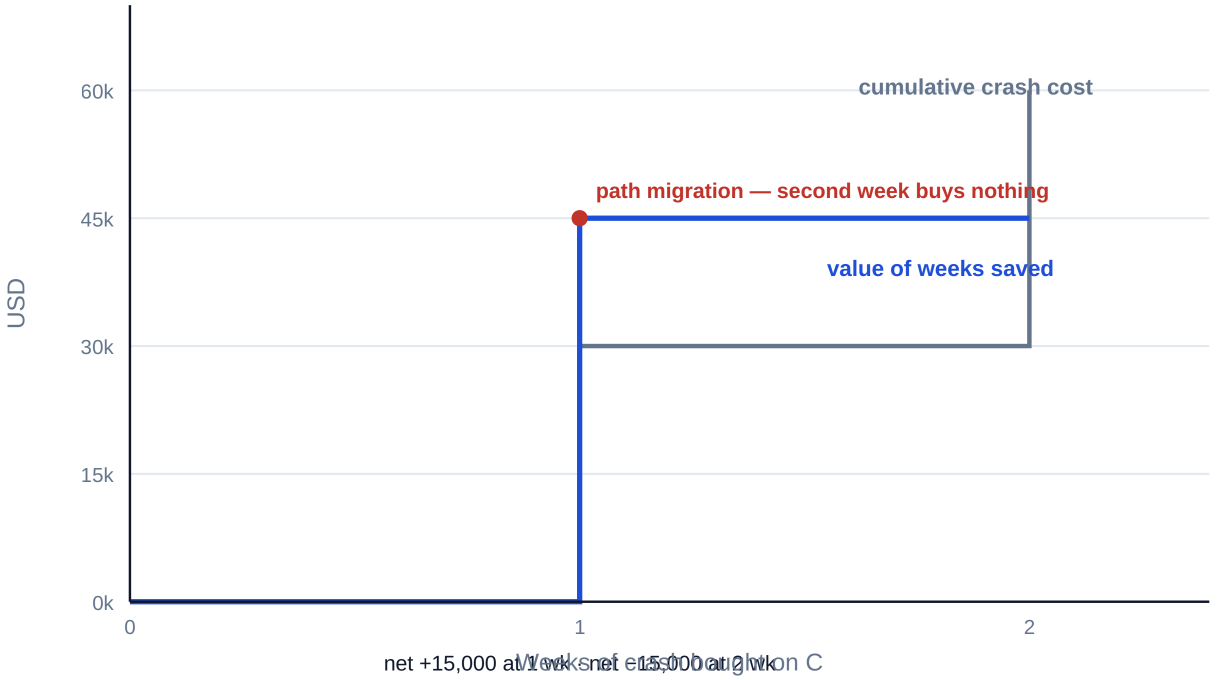

Fig 6.4.1 — The economics of crashing Auriga's procurement

Recovery. When the network slips (self-check 6.2.3: D at 10 weeks → 27), recovery options rank by cost-of-time: re-sequencing and logic re-choice (cheapest — KA 6.1's chosen logic); float harvesting from non-critical paths (G's 8 weeks fund nothing on the new critical path, but its resources might); crashing the new critical path (now through D); fast-tracking E against D's tail with an explicit rework risk (Domain 8's risk register prices it); and scope or acceptance re-negotiation (Domain 5) as the honest last resort. A **recovery plan** states the target date, the moves, their costs and risks, and the decision authority spending them — the template is toolkit 6.T.2.

6.4.3 Forecasting and scenario analysis

Three-point estimates. Uncertain durations are ranges, not points. The PERT expected duration and spread for an activity estimated optimistic `o`, most-likely `m`, pessimistic `p`:

$$t_e = (o + 4m + p) / 6 \qquad \sigma = (p - o) / 6$$

WORKED EXAMPLE

Worked example 6.4.3 — Auriga's installation risk.

1. **Setup.** Installation E is estimated $o = 4$, $m = 5$, $p = 12$ weeks (the long tail: a legacy-system integration problem discovered late).
2. **Formula.** $t_e = (o + 4m + p)/6$; $\sigma = (p - o)/6$.
3. **Substitution.** $t_e = (4 + 20 + 12)/6 = 36/6$; $\sigma = (12 - 4)/6 = 8/6$.
4. **Result.** t_e = **6.0 weeks** (not 5); σ = **1.33 weeks**.
5. **Interpretation.** The deterministic network used 5 weeks; the risk-weighted expectation is 6 — the skewed tail alone adds a week to the honest forecast, pushing expected completion toward 26 weeks before any risk event "happens". Deterministic dates are the mode, not the mean; boards deserve the mean and the spread. Full quantitative schedule risk analysis — correlating activities, simulating paths, criticality indices — is Domain 8 (KA 8.2), built on exactly this input.

Scenario analysis. Between the single date and the full simulation sits the leader's workhorse: three coherent scenarios (base / threat / opportunity), each a complete re-run of the passes under stated assumptions, each with owners for the assumptions that differ. Scenarios are cheap, auditable, and — because each is a real network — they expose path migration that a percentage-confidence number hides.

The earned-schedule bridge. Once cost joins schedule (Domain 7), progress data yields a time-based forecast: **earned schedule** ES asks when the value now earned was planned to have been earned, and $SPI(t) = ES / AT$ (actual time) measures schedule efficiency in time units. If Auriga's week-22 status shows work that the baseline expected by week 20, $SPI(t) = 20/22 = 0.91$ — the programme is running at 91 % of planned tempo, a signal the currency-based SPI famously loses late in a project. Domain 7 (KA 7.3) builds the full machinery; it is flagged here because schedule forecasting belongs to whoever owns the network, not only to the cost engineer.

AI in this KA

Machine forecasting earns trust in exactly one way: calibration. A model that ingests progress data and produces completion forecasts must show its record — forecast vs actual across past periods — before its numbers enter a board pack (Domain 14, KA 14.3's verification workflow). The leader's questions are the auditor's: what data trained it, what does it do when the data thins, who re-ran its arithmetic, and what would make it wrong? Fluent confidence without a calibration record is the machine version of the site manager who is "sure it'll be fine".

KEY TERMS — KA 6.4

Term	Meaning
Crashing / fast-tracking	Buying duration with money / with overlap risk.
Path migration	The critical path moving as durations change; why passes are re-run.
Recovery plan	Target date, ranked moves, costs, risks, decision authority.
Three-point / PERT estimate	$t_e = (o+4m+p)/6$, $\sigma = (p-o)/6$.
Scenario analysis	Complete network re-runs under coherent stated assumptions.
Integration milestone	The CPM node where an agile stream's delivery enters the network.

SAMPLE MCQS — KA 6.4

MCQ 6.4-A `[6.4.2 · Application]` Saving a week is worth USD 45,000; crashing the critical activity costs USD 30,000 per week for up to two weeks; after one week of crashing, a parallel path becomes co-critical. The value-maximising decision is: - A. crash two weeks (net +USD 30,000) - B. crash one week (net +USD 15,000) - C. crash nothing (avoid all cost) - D. crash two weeks and fast-track the parallel path at no cost

Rationale: Week one converts to a project week: +15,000. Week two is absorbed by the co-critical path: −30,000. A prices weeks that don't materialise; C leaves +15,000 unclaimed; D invents a free fast-track — overlap always carries rework risk.

MCQ 6.4-B `[6.4.3 · Application]` An activity is estimated o = 4, m = 5, p = 12 weeks. Its PERT expected duration is: - A. 5.0 weeks - B. 7.0 weeks - C. 6.0 weeks - D. 6.5 weeks

Rationale: `(4 + 4×5 + 12)/6 = 36/6 = 6.0`. A is the most-likely value (the mode); B is the unweighted mean of the three points; D miscounts the weighting.

MCQ 6.4-C `[6.4.1 · Analysis]` A hybrid programme needs a software module from an agile team for an integration milestone. The schedule-sound way to join the two worlds is: - A. impose the CPM date on the team as a sprint deadline - B. enter the team's velocity-based forecast as a ranged duration and read the latest acceptable delivery from the backward pass - C. exclude the module from the network since agile work cannot be scheduled - D. convert the agile team to predictive planning for the integration period

Rationale: B translates in both directions — evidence-based forecast in, latest-start requirement out. A dictates without evidence; C leaves the network blind at a convergence point; D destroys the team's delivery system to decorate the network.

MCQ 6.4-F `[6.4.3 · Application]` An activity is estimated optimistic 3, most-likely 4, pessimistic 8 weeks. Its PERT expected duration and standard deviation are: - A. $t_e = 4.5$, $\sigma = 0.83$ - B. $t_e = 4.0$, $\sigma = 0.83$ - C. $t_e = 5.0$, $\sigma = 1.67$ - D. $t_e = 4.5$, $\sigma = 5.0$

Rationale: $t_e = (3 + 16 + 8)/6 = 4.5$; $\sigma = (8 - 3)/6 = 0.83$. B reports the mode as the mean; C is the unweighted three-point average with a doubled spread; D confuses σ with the pessimistic-minus-optimistic range.

MCQ 6.4-D `[6.4.3 · Application]` At week 22, a programme has earned the value its baseline planned to earn by week 20. Its time-based schedule performance index `SPI(t)` is: - A. 1.10 - B. 0.91 - C. 0.80 - D. 20.0

Rationale: `SPI(t) = ES/AT = 20/22 = 0.91` — the programme delivers at 91 % of planned tempo. A inverts the ratio; C subtracts the two weeks from the wrong base; D reports earned schedule itself, not the index.

MCQ 6.4-E `[6.4.3 · Analysis]` Compared with quoting "78 % confidence of the date", giving the board three fully re-run schedule scenarios is stronger because: - A. three numbers always beat one - B. each scenario is an auditable network with named assumptions, showing how the date moves and where the path migrates - C. percentages are always statistically invalid - D. scenarios eliminate the need for risk analysis

Rationale: The scenario's power is auditability and mechanism — assumptions with owners, visible path migration. A is numerology; C overclaims (a calibrated percentage is legitimate, Domain 8); D reverses the relationship — scenarios are inputs to risk analysis, not substitutes.

SELF-CHECK — KA 6.4

1. Why does compression show diminishing returns? — Each crash promotes the next-longest path; saved weeks stop converting to project weeks (path migration).
2. What makes a scenario more useful to a board than a confidence percentage? — It is a complete, auditable network with named assumptions — it shows how the date moves, not just that it might.
3. When may a machine forecast enter a board pack? — With a calibration record, verified arithmetic, stated data lineage and a named human owner.

Advanced topics — Domain 6

6.A.1 The drum: constraint-aware flow

Where one resource is the system's constraint (a single commissioning team, one test rig), throughput scheduling subordinates everything to that drum: buffer it, never starve it, and let non-constraint efficiency go. This reframes utilisation worship — a 100 %-busy non-constraint builds inventory, not progress. In schedule terms: protect the constraint's feed path with deliberate float placement rather than spreading buffer evenly.

6.A.2 Schedule baselines and the update cycle

The control schedule lives on a cycle: status (actuals and remaining durations in), re-run passes, variance vs baseline, forecast, and — rarely, formally — re-baseline under Domain 4's change control. Two integrity rules carried from the family's controls tradition: never edit history (actualised dates are records), and never re-baseline to hide variance — a re-baseline is a governance decision with an audit trail, not a cosmetic one.

6.A.3 PDM edge cases: float under SS and FF links

Precedence-diagram (PDM) links change how float behaves, and two edge cases bite reviewers. **(1) The SS-linked short successor.** P (10 weeks) `SS+2` Q (3 weeks): Q can run weeks 2–5, finishing seven weeks before its predecessor — legal, and correct if the dependency truly binds only the starts; but if Q also needs P's output to finish, the network is missing an FF link, and Q's float is fiction. Paired `SS+FF` links express "starts together, finishes together" honestly. **(2) Duration-driven float.** Under an FF link the successor's float depends on its own duration: lengthen Q and its `LS` moves earlier — planners who "add duration for safety" on FF-linked activities silently destroy their float. The audit habit: on every SS/FF link, ask which end of each activity the logic genuinely binds, and whether the unbound end needs its own link. A network is only as honest as its least-considered link type.

6.A.4 Negative float and the honest schedule

A constraint-bound network can compute `TF < 0`: the plan, as constrained, is late already. Negative float is information — the size of the recovery problem — and suppressing it (deleting constraints, shortening durations by fiat) is the scheduling equivalent of cooking the books. The professional response mirrors KA 6.4.2: quantify it, trace the binding path, price the recovery options, and escalate the decision to whoever owns the trade (Domain 3's escalation design).

WORKED EXAMPLE

Worked example 6.A.4 — buying back two weeks of negative float.

1. **Setup.** A newly imposed outage constraint requires Auriga to finish by **week 23**; logic says 25 — so TF = −2 on A–B–C–E–F. Available buys: crash C at USD 30,000/week (max 2); crash E at USD 55,000/week (max 1). Find the cheapest feasible recovery to week 23.
2. **Formula.** Recover week by week on the currently binding path(s), re-running the passes after each buy (path migration, 6.4.2).
3. **Substitution.** Week one: crash C by 1 → duration 24, but B–D–E–F is now co-critical at 24. Week two must shorten both paths: the only shared activity available is E → crash E by 1 (USD 55,000) → duration 23. (C's second week alone would leave B–D–E–F at 24 — spent money, no schedule.)
4. **Result.** Feasible recovery: **crash C ×1 + crash E ×1 = USD 85,000**, finishing week 23.
5. **Interpretation.** Negative float is bought back on the binding path as it migrates — the second-cheapest week on the original critical path (C's second week, USD 30,000) is worthless, while the dearest activity (E, on both paths) is the only week-two purchase that works. This is why recovery plans list options in re-run order, not in unit-cost order (toolkit 6.T.2).

Industry variations — Domain 6

The passes are universal; what counts as a schedule, and what discipline it needs, is sectoral:

- **Construction and EPC.** Deep L3/L4 hierarchies, thousand-activity networks, contractual float-ownership clauses (Domain 10) and delay-analysis protocols make float a legal quantity — the honesty settings of the Executive perspective are contract compliance here, not just culture.
- **Technology and product.** Hybrid is the default (KA 6.4.1): cadence-based streams joined to a thin predictive network at integration milestones; the commonest defect is a network so sparse that convergence points are invisible until they slip.
- **Shutdowns and turnarounds.** The network runs in hours; near-critical analysis widens to every path within minutes of critical, and the drum logic of 6.A.1 governs a single shared crane or permit desk. Compression economics (6.4.2) turn over in shifts, not weeks.
- **Pharmaceutical and regulated development.** Milestones are regulatory events with submission-window physics — genuine date constraints (6.3.2), modelled as logic, with the pin-and-squeeze temptation at its most dangerous because windows are unforgiving.
- **Public programmes.** Announced dates precede networks (Case study B); the leader's protection — milestones only after the backward pass — is hardest exactly where it matters most, and negative-float honesty (6.A.4) is the difference between an early re-plan and a public failure.

Case study — Domain 6: recovering Auriga (utilities / technology)

Situation. End of week 13. Civil and cabling works D — planned at 7 weeks from week 8 — hits contaminated-ground remediation; the discipline lead's honest re-estimate takes D to 10 weeks (finish week 18). The client's outage window for final commissioning opens at week 25 and closing the window costs USD 45,000 per week of delay. The team re-runs the passes.

Analysis. With D = 10: E starts week 18 (D now later than C's week 16), F finishes week 27 — **two weeks into the penalty window**, and the critical path has migrated to **A–B–D–E–F**. Procurement C, critical all project, now carries TF = 2. G still holds float.

The recovery decision. Options priced by the team: (1) crash the old critical path — worthless, C is no longer binding; (2) crash D's remediation with a second civil crew — USD 35,000 for one week; (3) fast-track E against D's tail — start installation in the uncontaminated sections at week 16, accepted rework risk assessed at 20 % × USD 60,000 = USD 12,000 expected cost, one further week saved; (4) do nothing — 2 × 45,000 = USD 90,000 penalty exposure. The leader takes (2) + (3): cost ≈ USD 47,000 expected against USD 90,000 — and finishes at week 25 with the window intact.

What the domain teaches here. Re-run the passes before spending a cent (the instinctive "crash procurement" would have bought nothing); price recovery in expected cost including risk; harvest overlap where the physics genuinely allows it; and report the new float positions to the client honestly — the schedule survived because the remediation estimate was honest a full five weeks before the window (Domain 11's reporting culture, applied to time).

Case study B — Domain 6: the milestone that was typed, not computed (public programme)

Situation. A government services programme publishes a go-live milestone eighteen months out, set in a ministerial announcement before the L3 schedule existed. When the network is finally built, the backward pass puts the milestone's required start for user-acceptance testing three weeks before the environment that testing needs can exist — TF = −3 from day one.

What happened. For six reporting cycles the programme's L1 view showed the milestone green: the date was pinned, and each slip downstream was absorbed by silently shortening the testing window — the classic pin-and-squeeze. In cycle seven an assurance review (Domain 3, KA 3.3) ran the passes without the pin, surfaced the negative float, and forced the choice the pin had deferred: descope the first release (Domain 5), or move the date. The minister moved the date — at a political cost several times what a computed date would have cost eighteen months earlier.

What the domain teaches here. A typed milestone is a promise without a plan. Negative float is not embarrassment to be formatted away; it is the earliest, cheapest warning the programme will ever get (6.A.4). And the leader's protection is procedural, not heroic: milestones enter public commitments only after the backward pass says they exist.

Executive perspective — Domain 6

What a project leader cannot delegate in this domain:

- **The logic sign-off.** Discipline leads own their links; the leader owns the assembled claim that this network is how the project actually works.
- **Float policy.** Who may spend float, in what increments, reported how — decided before the first progress update, not during the first crisis.
- **The honesty settings.** No pins doing logic's work; negative float reported the cycle it appears; re-baselining only through governance. Schedules fail morally before they fail mathematically.

- **The compression chequebook.** Crash and fast-track decisions are investments under uncertainty — the leader signs the expected-value arithmetic (6.4.2) and the risk acceptance (Domain 8), never a bare instruction to "go faster".
- **The translation duty.** Boards hear dates; teams live in passes, floats and scenarios. The leader is the honest converter between the two languages — both directions.

Calculation exercises — Domain 6

Exercise 6.1 Using the Auriga network, compute `ES`/`EF` for every activity and the project duration, showing the convergence at E. Solution. A 0–2 · B 2–8 · C 8–16 · D 8–15 · E max(16,15)=16–21 · F 21–25 · G 15–17. Duration **25 weeks**; E's start is set by C (16), not D (15) — the convergence rule takes the later predecessor. Common error: taking the earlier predecessor at a merge (giving E 15–20 and a false 24-week project).

Exercise 6.2 Complete the backward pass and state `TF` and `FF` for D and G. Solution. From `LF` = 25: F 21/25 · E 16/21 · C 8/16 · G 23/25 · D `LF` = min(E LS 16, G LS 23) = 16, `LS` = 9. **D: `TF` = 1, `FF` = min(16,15) − 15 = 0. G: `TF` = 8, `FF` = 25 − 17 = 8.** Common error: computing D's `FF` against only E (giving 1) — free float tests every successor.

Exercise 6.3 The client offers USD 45,000 per week saved; crashing C costs USD 30,000/week (max 2 weeks); crashing E costs USD 55,000/week (max 1 week). Find the value-maximising plan. Solution. Crash C by 1: 25→24, +15,000. Second week of C: co-critical B–D–E–F holds 24 — −30,000, reject. Crash E by 1 (E is on both paths): 24→23, `45,000 − 55,000 = −10,000`, reject. **Plan: crash C by one week; net +USD 15,000.** Common error: crashing E first because it is "on more paths" — path count doesn't change its negative unit economics.

Exercise 6.4 Activity durations: o = 6, m = 8, p = 16 weeks. Compute t_e and `σ`, and state the deterministic bias. Solution. t_e `= (6 + 32 + 16)/6 =` **9.0 weeks**; `σ = (16 − 6)/6 =` **1.67 weeks**. The deterministic plan at m = 8 understates the expectation by a full week — right-skewed tails always pull the mean above the mode.

Exercise 6.5 A programme's network shows `TF` = −2 on the binding path. List, in cost order, the recovery families and the governance step each requires. Solution. (1) Logic re-choice/re-sequencing — planner's proposal, leader's sign-off; (2) float and resource harvesting from non-binding paths — float-policy authority; (3) crashing the binding path — expected-value case, budget authority; (4) fast-tracking — risk acceptance recorded in the register; (5) scope/acceptance renegotiation — sponsor and change control. Reporting the −2 honestly precedes all five (6.A.4).

Practitioner's toolkit — Domain 6

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 6.T.1 — Schedule quality gate (run before trusting any network)

- [ ] Zero dangles; zero loops; every lag has a stated reason.

- [ ] Date constraints listed, each justified as genuine external physics — none doing logic's job.
- [ ] Durations owned by the executing discipline, estimate class stated (far waves: ranged).
- [ ] Passes re-run after every change; critical-path durations sum exactly to project duration.
- [ ] Near-critical paths (within 10 %) listed with their floats.
- [ ] Float report distinguishes TF and FF; float policy names who may spend it.
- [ ] Milestones computed, owned, done-defined; no typed dates.
- [ ] AI-drafted logic or forecasts marked, verified, and owned by a named human.

Toolkit 6.T.2 — Recovery plan (one page)

Slip statement (activity, cause, size, date detected) · re-run pass results (new duration, new critical path, float table) · options priced in expected cost (re-sequence / harvest / crash / fast-track / renegotiate), each with risk and owner · selected plan and its decision authority · revised commitments and the stakeholder notice list · review date.

Toolkit 6.T.3 — Milestone register

Per milestone: computed date and current TF · owner · done-definition (acceptance reference) · contractual status and counterparty · L3 logic trace · reporting treatment (internal / contractual / public) · change history.

Exam preparation — Domain 6

The calculation traps. Taking the earlier predecessor at a convergence (Exercise 6.1) · computing FF against one successor instead of all (Exercise 6.2) · quoting TF after it has been spent · crashing from the original network instead of re-running passes (path migration, 6.4.2) · treating the PERT mode as the mean (6.4.3) · reading a pinned milestone as a forecast · confusing project-level float with activity-level slack in reports.

Reflection questions. 1. Your programme's board pack shows every milestone green. What three questions establish whether that is information or formatting? (Computed or typed? Passes re-run this cycle? Where is the float table?) 2. A subcontractor asks for "the float" on their package. What must you settle before answering? (Which float — *TF* or *FF*; the float policy; contractual float ownership — Domain 10.) 3. When did you last see money spent on compression that the passes would have shown was worthless — and which governance step was missing? (6.4.2; toolkit 6.T.2.)

Domain 6 summary

This domain turned "when will it finish?" from an opinion into an instrument. Planning levels give each audience a true view of one logic; dependency discipline makes the network a model rather than a drawing; the forward and backward passes yield the dates, the two floats and the critical path — with the master project showing why free float can vanish while total float survives, and why convergence points compound slippage. Resources, constraints, milestones and rolling waves connect the network to the real world's crews, windows and horizons. Delivery flow closes the loop: compression priced by expected value and re-run passes, recovery ranked from re-choice to renegotiation, PERT and scenarios replacing single dates with honest ranges, and agile streams joined to the network through ranged integration milestones. The leadership spine throughout: schedules fail morally before they fail mathematically — computed dates, visible float, reported negative float, and machine output verified by named humans are what make a programme's word worth something.

07

DOMAIN 7

Cost, Resources and Commercial Awareness (quantitative flagship)

Group: Delivering the work (Domain 7 of 6 in Part Two — the cost counterpart to Domain 6's schedule). **Target:** ~78 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is this book's home of the earned-value symbols — `PV`, `EV`, `AC`, `BAC`, `CV`, `SV`, `CPI`, `SPI`, `EAC`, `ETC`, `VAC`, `TCPI` — carried unchanged from the PCI family master table. Note the family notation-clash rule: `PV` here is **Planned Value**; present value is written `PV(x)` or in words (PFL-AI, Domain 3). British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

A leader who can defend a date but not a number is half-equipped. Domain 6 built the schedule; this domain builds the money that pays for it and the commercial arrangements that bind other people to deliver it. It starts with estimating and budgeting — how a credible number is constructed and what its accuracy claim actually means (KA 7.1); establishes the cost baseline and the forecasting that keeps it honest (KA 7.2); builds **earned value** in full, because it is the only technique that answers cost and schedule performance in one integrated language (KA 7.3); and closes with the commercial awareness a leader cannot delegate — resource economics, procurement strategy, contract models and cash flow (KA 7.4). Risk quantification deepens in Domain 8; contracts and supply networks get their own treatment in Domain 10. What belongs here is the leader's own numeracy: enough to know when a forecast is arithmetic and when it is hope.

Learning objectives. After this domain a candidate can: select an estimating method appropriate to the available definition and state its accuracy class; build a three-point cost estimate; assemble a cost baseline separating contingency from management reserve; compute and interpret the full earned-value set (`CV`, `SV`, `CPI`, `SPI`); forecast with each `EAC` method and choose the one matching the variance's cause; compute `VAC` and `TCPI` and explain what `TCPI` demands of the remaining work; read resource economics through blended rates; distinguish the main contract models by who carries cost risk; compute a cost-incentive fee and the point of total assumption; explain why cash flow and profit differ on a project; and govern AI-produced cost forecasts under the family verification rule.

The master worked project. Project Auriga continues from Domain 6 — the 25-week control-systems upgrade for a regional utility — now with money attached. Its approved cost baseline is `BAC` **= USD 4,000,000**. At the **data date, end of week 13**, the baseline says `PV` **= USD 2,080,000**; measurement gives `EV` **= USD 1,920,000** and `AC` **= USD 2,120,000**. Every calculation in KA 7.2–7.3 uses these four numbers.

KNOWLEDGE AREA 7.1

Estimating and budgeting

Topics: 7.1.1 estimating methods and accuracy classes · 7.1.2 three-point estimating · 7.1.3 from estimate to budget.

7.1.1 Estimating methods and accuracy classes

The principle. An estimate's accuracy is governed by how well the work is defined, not by how much effort went into the arithmetic. The three standard methods trade definition against speed:

Method	How it works	Needs	Typical use
Analogous (top-down)	Scale a comparable past project	A true analogue and a scaling basis	Concept, screening (Domain 2)
Parametric	Cost = rate × parameter (per metre, per point, per kW)	A calibrated rate from real history	Early design, repetitive work
Bottom-up	Estimate each work package, then sum	A WBS at package level (Domain 4)	Baseline setting, control

Accuracy classes. Mature cost practice publishes an estimate's class alongside its number — a range that narrows as definition matures (the AACE Total Cost Management framework's class-5-to-class-1 progression is the reference treatment, described here in this book's own words). The professional discipline is simple and widely broken: **an estimate is never a single number without a range and a class**. "USD 4 million" is not an estimate; "USD 4 million, −15 %/+30 %, at a class consistent with 30 % design completion" is.

Common pitfall — precision mistaken for accuracy. A bottom-up estimate summing 400 packages to `USD 4,183,662` looks authoritative and is no more accurate than its worst assumption. Leaders should be more suspicious of an over-precise number than a rounded one.

7.1.2 Three-point estimating

Cost, like duration (Domain 6, KA 6.4.3), is a distribution. The same PERT weighting applies:

$$C_e = (o + 4m + p) / 6 \qquad \sigma = (p - o) / 6$$

WORKED EXAMPLE

Worked example 7.1.2 — Auriga's control-hardware package.

1. **Setup.** Procurement of control hardware is estimated **optimistic USD 680,000**, **most-likely USD 750,000**, **pessimistic USD 1,000,000** (the tail: a single-source controller with volatile lead-time pricing).
2. **Formula.** $C_e = (o + 4m + p)/6$; $\sigma = (p - o)/6$.
3. **Substitution.** $C_e = (680{,}000 + 3{,}000{,}000 + 1{,}000{,}000)/6 = 4{,}680{,}000/6$; $\sigma = 320{,}000/6$.
4. **Result.** C_e = **USD 780,000**; $\sigma \approx$ **USD 53,333**.
5. **Interpretation.** The most-likely figure is 750,000, but the expected cost is 780,000 — the right-skewed tail adds USD 30,000 before anything goes wrong. Budgeting at the mode systematically under-funds a portfolio of such packages; the difference between mode and mean, summed across a project, is a large part of what contingency exists to cover (7.1.3, and Domain 8's quantification).

7.1.3 From estimate to budget

The structure. An estimate becomes a controllable budget through a deliberate hierarchy:

```
work-package estimates
  → control-account budgets            (WBS × organisation — the management-control points)
  → + contingency reserve              → the COST BASELINE (BAC): the PM's authority
  → + management reserve               → the total project funding requirement
```

Two rules carry the discipline. **Contingency is inside the baseline**, sized against identified risks (Domain 8) and spent by the project manager under a stated protocol. **Management reserve is outside the baseline**, held for unknown-unknowns, and released only by the sponsor or change authority (Domain 3's decision rights) — releasing it changes the baseline through change control (Domain 4, KA 4.4). Blurring the two is how projects appear to be "on budget" while consuming their own funding runway.

Time-phasing. The baseline is spread across the schedule to produce the **cumulative cost curve** (the S-curve) — and that curve is Planned Value (KA 7.3). A budget with no phasing cannot be measured against; the schedule (Domain 6) is therefore a precondition of cost control, not a parallel activity.

AI in this KA

Estimating is where AI assistance is most seductive and most in need of provenance. A model can produce a plausible parametric rate instantly, and it cannot tell you whether that rate came from comparable work, a different market, or nowhere at all. The governed workflow: **AI proposes** a method, a rate and a range; the estimator supplies the calibration evidence (which projects, which years, which escalation basis — the source discipline of the shared registry); the range and class are stated; and a named human owns the number. **AI proposes; the professional verifies, decides and remains accountable.** An estimate whose basis cannot be produced on request is not an estimate.

KEY TERMS — KA 7.1

Term	Meaning
Analogous / parametric / bottom-up	Scale an analogue · rate × parameter · sum the packages.
Accuracy class	The definition-linked range accompanying an estimate; never optional.
Control account	The WBS × organisation point where scope, budget and actuals integrate.
Contingency reserve	Inside the baseline; funds identified risks; PM-controlled.
Management reserve	Outside the baseline; unknown-unknowns; sponsor-controlled.
`BAC`	Budget at Completion — the time-phased cost baseline's total.

SAMPLE MCQS — KA 7.1

MCQ 7.1-A `[7.1.2 · Application]` A package is estimated o = 680,000, m = 750,000, p = 1,000,000. Its PERT expected cost is: - A. USD 750,000 - B. USD 780,000 - C. USD 810,000 - D. USD 840,000

Rationale: `(680,000 + 4 × 750,000 + 1,000,000)/6 = 780,000`. A is the mode; C is the unweighted three-point mean; D over-weights the pessimistic value.

MCQ 7.1-B `[7.1.3 · Analysis]` A project reports "on budget" while having consumed 60 % of its management reserve at 40 % complete. The correct reading is: - A. genuinely on budget — management reserve exists to be spent - B. the baseline is intact but the project's total funding is eroding faster than progress; the trend belongs in the next report to the sponsor - C. a baseline breach requiring immediate re-baselining - D. an accounting error, since management reserve is inside the baseline

Rationale: Management reserve sits outside the baseline (so D is wrong and A is technically true but misleading), and its release is a sponsor-level signal, not a project-level convenience. It is not yet a breach (C) — it is the early warning that precedes one.

MCQ 7.1-C `[7.1.1 · Recall]` Which statement about a bottom-up estimate summing to USD 4,183,662 is soundest? - A. its precision indicates high accuracy - B. its accuracy is bounded by its assumptions and definition maturity, and it must still carry a range and class - C. rounding it would reduce its accuracy - D. bottom-up estimates do not need accuracy classes

Rationale: Precision (digits) and accuracy (closeness to outturn) are independent; definition maturity governs the latter. Rounding changes no information (C), and every estimate carries a class (D).

SELF-CHECK — KA 7.1

1. Where do contingency and management reserve sit, and who spends each? — Contingency inside the baseline, PM-controlled; management reserve outside it, sponsor-controlled.
2. Why is the time-phased baseline a precondition for cost control? — Because the phased cumulative curve is Planned Value; without it there is nothing to measure performance against.
3. What two things must accompany every estimate? — A range and an accuracy class tied to definition maturity.

KNOWLEDGE AREA 7.2

The cost baseline, actuals and forecasting

Topics: 7.2.1 measuring actual cost · 7.2.2 the forecasting question · 7.2.3 baseline integrity.

7.2.1 Measuring actual cost

The principle. `AC` must cover the same work as `EV`, in the same period, or every index built from them is fiction. Three mechanics decide whether it does:

- **Accruals.** Work received but not yet invoiced belongs in this period's `AC`. Omit accruals and cost performance looks excellent until the invoices land — the commonest cause of a "sudden" overrun that was months old.
- **Commitment vs actual.** A purchase order is a commitment, not a cost. Both matter: `AC` drives performance, commitments drive the funding forecast. Confusing them double-counts or hides money.
- **Open-commitment hygiene.** Stale purchase orders left open inflate the forecast; cleansing them is a standing month-end task.

7.2.2 The forecasting question

A forecast answers one question — what will this cost when it is done? — and the honest answer depends entirely on **why** the current variance exists. That judgment is the leader's; the arithmetic is KA 7.3.3's `EAC` family. The rule to internalise now: **a forecast is a statement about the remaining work, not an extrapolation ritual.** A variance caused by a closed, non-recurring event says nothing about what is left; a variance caused by a systemic productivity shortfall says everything.

7.2.3 Baseline integrity

The baseline's authority rests on its stability. Four standing rules: budgets of completed or open work packages are never retrospectively adjusted (except to correct an authorised error); re-baselining happens only through change control, with an audit trail (Domain 4, KA 4.3); transfers between control accounts are logged, not silent; and the baseline never absorbs a variance by moving budget to where the money went. A baseline edited to match reality has stopped measuring anything — the cost analogue of Domain 6's pinned milestone.

KEY TERMS — KA 7.2

Term	Meaning
`AC`	Actual Cost of the work performed, including period accruals.
Accrual	Cost of work received but not yet invoiced, recognised in the period.
Commitment	A contractual obligation (e.g. a PO); a funding fact, not yet a cost.
Re-baselining	A governed change to the baseline; never a variance-hiding device.

SAMPLE MCQS — KA 7.2

MCQ 7.2-A `[7.2.1 · Analysis]` A project's `CPI` has read 1.02 for four months; then one month it drops to 0.91 with no change in productivity. The likeliest explanation is: - A. genuine sudden inefficiency - B. accruals were not being recognised, so earlier AC understated cost and this month absorbed the catch-up - C. the baseline was too generous - D. EV was over-claimed this month

Rationale: A step change in CPI without an operational change points at measurement, and missing accruals are the classic cause — earlier periods flattered, one period punished. D would raise, not lower, the earlier readings' credibility, and C would show as a stable, not stepped, pattern.

MCQ 7.2-B [7.2.3 · Recall] Which action preserves baseline integrity? - A. moving budget from an underspent control account to cover an overspend, without record - B. re-baselining through change control with an audit trail when scope genuinely changes - C. adjusting a completed package's budget to match its actual cost - D. reducing remaining budgets so the total still equals BAC

Rationale: Only governed change preserves the measurement. A is a silent transfer, C is retrospective adjustment, D is the classic "make the numbers add up" manoeuvre — each destroys comparability.

SELF-CHECK — KA 7.2

1. Why must *AC* include accruals? — So it covers the same work as EV in the same period; otherwise every index is misstated.
2. Is a purchase order a cost? — No: a commitment. It informs the funding forecast, not performance.
3. What single question does a forecast answer? — What the remaining work will cost, given why the variance exists.

KNOWLEDGE AREA 7.3

Earned value: measurement, variances and forecasting

Topics: 7.3.1 the three measures · 7.3.2 variances and indices · 7.3.3 the *EAC* family · 7.3.4 *VAC* and *TCPI*.

7.3.1 The three measures

Definitions. All three are expressed in the same budget currency so they are directly comparable:

- **Planned Value (PV)** — the budgeted cost of the work scheduled by the data date: the time-phased baseline of KA 7.1.3.
- **Earned Value (EV)** — the budgeted cost of the work actually performed: physical progress **valued at the budget rate**, never at what it cost.
- **Actual Cost (AC)** — the cost actually incurred for that work, accruals included (KA 7.2.1).

The single most important conceptual point: EV is measured at budget. That is what lets EV be compared with PV (both at budget → schedule progress) and with AC (budget vs actual for the same work → cost efficiency). Confusing "value earned" with "cost incurred" collapses the method.

Earning rules. How EV is claimed decides how much it can be gamed: **0/100** (nothing until complete — objective, good for short packages); **50/50**; **percent complete** (needs an objective basis); **units completed** (best where output is countable); **weighted milestones**; and **level of effort**, where EV is set equal to PV by the calendar, so it can never show a schedule variance. Level of effort dilutes whatever it is mixed into — formal practice segregates it and caps its share of a control account.

7.3.2 Variances and indices

```
CV = EV − AC        SV = EV − PV
CPI = EV / AC       SPI = EV / PV
```

WORKED EXAMPLE

Worked example 7.3.2 — Auriga at week 13.

1. **Setup.** `BAC` = 4,000,000; at the data date `PV` = 2,080,000, `EV` = 1,920,000, `AC` = 2,120,000.
2. **Formula.** As above; variances in currency, indices as ratios (two decimals).
3. **Substitution.** `CV = 1,920,000 − 2,120,000`; `SV = 1,920,000 − 2,080,000`; `CPI = 1,920,000/2,120,000`; `SPI = 1,920,000/2,080,000`.
4. **Result.** `CV` = **(USD 200,000)** · `SV` = **(USD 160,000)** · `CPI` = **0.91** · `SPI` = **0.92**. Auriga is **48.0 % complete** (`EV/BAC`) having spent **53.0 %** of budget (`AC/BAC`).
5. **Interpretation.** Behind and over: for every dollar spent the project is producing 91 cents of budgeted work, and it has delivered 92 % of what it planned to by now. The percent-complete versus percent-spent pair (48 vs 53) is the same story in the form a board absorbs fastest. Note this is the cost view of the very slippage Domain 6's case study recovered on the schedule side — the two domains are describing one project.

Fig 7.3.1 — Auriga's earned-value S-curves at week 13

7.3.3 The EAC family — forecasting

```
ETC = the remaining work's forecast cost          EAC = AC + ETC
(a) EAC = AC + (BAC − EV)                     remaining work at the BUDGETED rate
(b) EAC = BAC / CPI                           current cost efficiency PERSISTS
(c) EAC = AC + (BAC − EV) / (CPI × SPI)       cost AND schedule pressure compound
(d) EAC = AC + bottom-up ETC                  re-estimate the remainder from scratch
```

WORKED EXAMPLE

Worked example 7.3.3 — the same project, four futures.

1. **Setup.** Auriga's week-13 figures, forecast to completion by each method.
2. **Formula.** (a)–(c) above; `CPI` = 0.905660, `SPI` = 0.923077 at full precision.
3. **Substitution.** (a) `2,120,000 + (4,000,000 − 1,920,000)`; (b) `4,000,000 / 0.905660`; (c) `2,120,000 + 2,080,000/(0.905660 × 0.923077)`.
4. **Result.** (a) **USD 4,200,000** · (b) **USD 4,416,667** · (c) **USD 4,608,056**.
5. **Interpretation.** The same four measured numbers support forecasts spanning **USD 408,056** — over 10 % of `BAC`. The spread is not uncertainty in the arithmetic; it is the assumption about the remaining work, made explicit. Choosing among them is the leader's judgment, and it must be stated: if Auriga's overrun was the contaminated-ground remediation of Domain 6 — a discrete, now-closed event — method (a) is right and the others over-forecast. If it reflects a systemic productivity shortfall that will persist, (b). If schedule pressure is now driving overtime and expediting, (c). A forecast presented without its assumption named is not a forecast; it is a number.

7.3.4 VAC and TCPI

```
VAC = BAC − EAC                              the forecast overrun/underrun
TCPI = (BAC − EV) / (BAC − AC)               efficiency the REMAINING work must achieve to hit BAC
TCPI = (BAC − EV) / (EAC − AC)               …to hit a revised EAC
```

WORKED EXAMPLE

Worked example 7.3.4 — what recovery would actually require.

1. **Setup.** Auriga's week-13 figures; test recovery to `BAC`, and to the method-(b) forecast.
2. **Formula.** `VAC = BAC − EAC`; `TCPI` as above.
3. **Substitution.** `VAC(a) = 4,000,000 − 4,200,000`; `VAC(b) = 4,000,000 − 4,416,667`. `TCPI(BAC) = 2,080,000 / (4,000,000 − 2,120,000) = 2,080,000/1,880,000`. `TCPI(EAC b) = 2,080,000 / (4,416,667 − 2,120,000)`.
4. **Result.** `VAC(a)` = **(USD 200,000)** · `VAC(b)` = **(USD 416,667)**. `TCPI` to `BAC` = **1.11**; `TCPI` to the (b) forecast = **0.91**.
5. **Interpretation.** To finish at budget, the remaining 52 % of work must run at **1.11** — an 11 % efficiency gain by a team currently achieving 0.91. That is a 22-point swing, and the honest question in the room is what specifically would deliver it. Meanwhile `TCPI` to the (b) forecast is 0.91 — exactly today's `CPI`, which is the arithmetic identity worth internalising: **forecasting with `BAC/CPI` is precisely the assumption that nothing changes.** `TCPI` is the reality check on every recovery promise: a required index far above demonstrated performance is a plan that needs a mechanism, not encouragement.

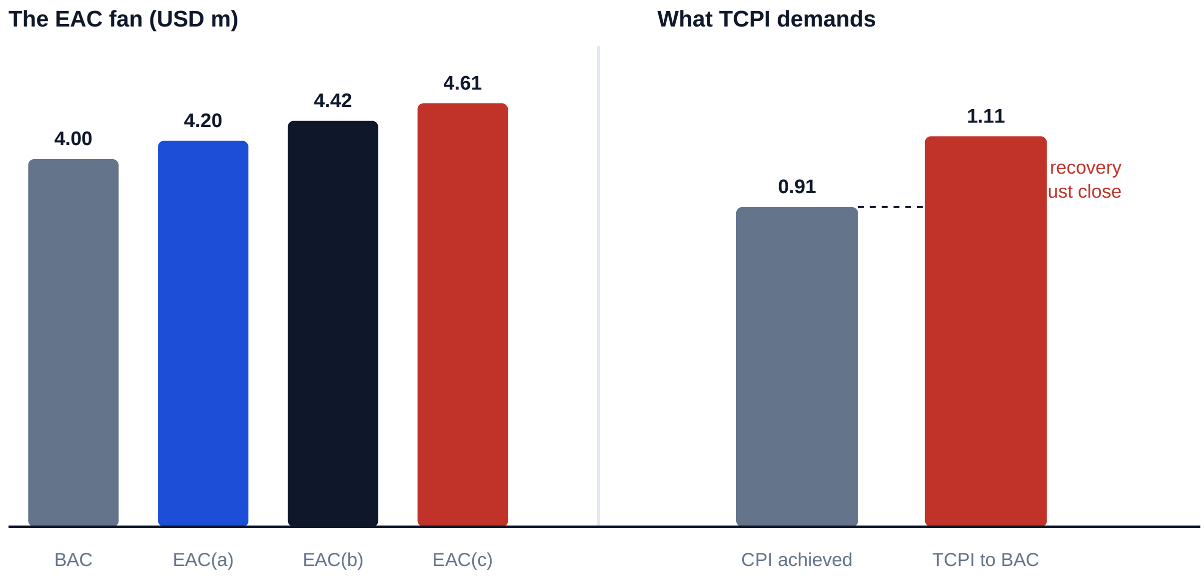

Fig 7.3.2 — The `EAC` fan and what `TCPI` demands

AI in this KA

Earned value is arithmetic on four numbers, so machine computation is safe and instant verification is cheap — which makes unverified AI output indefensible here rather than merely risky. The deterministic checks: `EV/BAC` must equal the claimed percent complete; `CV = EV − AC` and `CPI = EV/AC` recomputed independently; `TCPI` to `BAC/CPI` must equal `CPI` (7.3.4's identity); and every `EAC` must state its assumption. Where AI forecasts from trend data, the calibration rule of Domain 6 (KA 6.4) applies unchanged: show the record of past forecasts against outturn before the number enters a board pack, and name the human who owns it.

KEY TERMS — KA 7.3

Term	Meaning
`PV` `EV` `AC`	Budgeted cost of work scheduled · performed · the cost actually incurred.
`CV` `SV`	`EV − AC` · `EV − PV`, in currency.
`CPI` `SPI`	`EV/AC` · `EV/PV`, as ratios.
`EAC` `ETC`	Estimate at / to complete; `EAC = AC + ETC`.
`VAC`	`BAC − EAC` — forecast variance at completion.
`TCPI`	Efficiency the remaining work must achieve to hit a stated target.
Level of effort	Earning by calendar (`EV ≡ PV`); can never show schedule variance.

SAMPLE MCQS — KA 7.3

MCQ 7.3-A `[7.3.2 · Application]` `BAC` 4,000,000; `PV` 2,080,000; `EV` 1,920,000; `AC` 2,120,000. `CPI` and `SPI` are: - A. 0.91 and 0.92 - B. 1.10 and 1.08 - C. 0.92 and 0.91 - D. 0.91 and 1.02

Rationale: `CPI = 1,920,000/2,120,000 = 0.91`; `SPI = 1,920,000/2,080,000 = 0.92`. B inverts both ratios; C swaps them (dividing `EV` by the wrong denominator); D miscomputes `SPI` against `BAC`-derived progress.

MCQ 7.3-B `[7.3.3 · Analysis]` The overrun was caused by a one-off ground-remediation event, now closed, and the remaining work is expected to run to budget. The appropriate `EAC` is: - A. `AC + (BAC − EV)` = 4,200,000 - B. `BAC/CPI` = 4,416,667 - C. `AC + (BAC − EV)/(CPI × SPI)` = 4,608,056 - D. `BAC` = 4,000,000

Rationale: A discrete, closed cause makes the variance **atypical**, so remaining work is forecast at the budgeted rate. B assumes the inefficiency persists and C that it compounds with schedule pressure — both contradict the stated cause. D ignores money already spent above budget.

MCQ 7.3-C `[7.3.4 · Application]` With `BAC` 4,000,000, `EV` 1,920,000 and `AC` 2,120,000, the `TCPI` required to complete at `BAC` is: - A. 0.91 - B. 1.00 - C. 1.11 - D. 1.21

Rationale: `(4,000,000 − 1,920,000)/(4,000,000 − 2,120,000) = 2,080,000/1,880,000 = 1.11`. A is the demonstrated `CPI`; B assumes recovery needs only par performance; D uses `PV` in place of `AC` in the denominator.

MCQ 7.3-D `[7.3.1 · Analysis]` A control account is 70 % level-of-effort by budget. Its reported `SPI` of 1.00 most likely means: - A. the account is exactly on schedule - B. little about schedule: level of effort earns by the calendar, so `EV ≡ PV` for most of the account regardless of progress - C. the discrete work is ahead, offsetting a delay - D. the earning rules were misapplied

Rationale: LOE sets `EV` equal to `PV` by construction, so a heavily-LOE account reads 1.00 whatever happens — which is why practice segregates and caps it. C invents an offset the data cannot show; the rules may have been applied entirely correctly (D), and that is the problem.

SELF-CHECK — KA 7.3

1. Why is *`EV`* measured at budget, not at cost? — So it is comparable with `PV` (schedule) and with `AC` (efficiency); at cost it would collapse into `AC`.
2. What does *`TCPI`* = 1.11 against *`CPI`* = 0.91 tell a leader? — Recovery to budget requires a 22-point efficiency swing; the plan needs a named mechanism, not optimism.

3. State the identity linking *TCPI* and *BAC/CPI*. — TCPI to an EAC of BAC/CPI equals the current CPI: that forecast is the assumption that nothing changes.

KNOWLEDGE AREA 7.4

Resource economics, procurement strategy and cash

Topics: 7.4.1 resource economics and blended rates · 7.4.2 contract models and cost risk · 7.4.3 incentive fees and the point of total assumption · 7.4.4 cash flow versus profit.

7.4.1 Resource economics and blended rates

Most project cost is people. A leader who reasons in headcount rather than **cost per unit of capability** will mis-price every option in Domain 6's compression decisions.

WORKED EXAMPLE

Worked example 7.4.1 — Auriga's engineering blended rate.

1. **Setup.** The engineering pool is 40 technicians at USD 95/hour, 25 engineers at USD 140/hour, 15 senior specialists at USD 210/hour.
2. **Formula.** Blended rate = Σ(count × rate) / Σ(count).
3. **Substitution.** `(40×95 + 25×140 + 15×210) / 80 = (3,800 + 3,500 + 3,150)/80 = 10,450/80`.
4. **Result. USD 130.63 per hour** (130.625 exactly; ≈ SAR 490 indicatively).
5. **Interpretation.** The blended rate makes options commensurable: a week of schedule compression staffed from this pool costs a computable amount, and shifting the mix changes the price as much as changing the headcount — adding five specialists moves the blend more than adding five technicians. This is the number that turns Domain 6's "crash it" instruction into an expected-value decision.

7.4.2 Contract models and cost risk

Every contract model is an answer to one question: **who carries cost risk?**

Model	Buyer pays	Cost risk sits with	Suits
Firm fixed price (FFP)	An agreed price, whatever it costs	**Seller**	Well-defined scope; priced-in risk premium
Fixed price incentive (FPIF)	Target cost/fee with a share ratio, capped by a ceiling	**Shared, then seller above the ceiling**	Definable scope with real uncertainty
Cost plus fixed fee (CPFF)	Allowable cost + a fixed fee	**Buyer**	Development, unclear scope
Cost plus incentive fee (CPIF)	Allowable cost + a fee varying with performance	**Shared**	Scope where effort is uncertain but effort quality matters
Time and materials (T&M)	Rates × quantities	**Buyer**	Staff augmentation, short or open-ended work

The leader's discipline: **risk transferred is risk priced.** An FFP contract for ill-defined scope does not remove the risk — it converts it into a premium plus a claims exposure when the scope moves (Domain 10, KA 10.4). Conversely a cost-plus contract on well-defined work pays for uncertainty that no longer exists.

7.4.3 Incentive fees and the point of total assumption

The mechanics. Under an incentive contract, buyer and seller agree a **target cost**, a **target fee** and a **share ratio** (buyer/seller) for over- and under-runs, with a **ceiling price** capping the buyer's exposure.

WORKED EXAMPLE

Worked example 7.4.3 — the incentive that stops incentivising.

1. **Setup.** Auriga's installation subcontract: target cost **USD 2,000,000**, target fee **USD 150,000**, share ratio **70/30** (buyer 70 %, seller 30 %), ceiling price **USD 2,450,000**. The seller finishes at an actual cost of **USD 2,300,000**.
2. **Formula.** Fee = target fee − (overrun × seller share). Buyer pays = actual cost + fee (subject to the ceiling). Point of total assumption `PTA = target cost + (ceiling − target price)/buyer share`, where target price = target cost + target fee.
3. **Substitution.** Overrun `2,300,000 − 2,000,000 = 300,000`; fee `150,000 − 300,000 × 0.30 = 150,000 − 90,000`. Buyer pays `2,300,000 + 60,000`. Target price `2,150,000`; `PTA = 2,000,000 + (2,450,000 − 2,150,000)/0.70`.
4. **Result.** Fee **USD 60,000** (down from 150,000); buyer pays **USD 2,360,000**. `PTA` **= USD 2,428,571.43** — and at that cost the buyer pays exactly the USD 2,450,000 ceiling.
5. **Interpretation.** The share ratio does its job up to the `PTA`: both parties lose money on overrun, so both want efficiency. **Above the `PTA` the seller bears 100 %** of further cost — which is the moment the incentive inverts: a seller heading past it has no financial reason to spend more on your project and every reason to argue that the extra cost is your scope change. Knowing where the `PTA` sits is therefore a **delivery** insight, not an accounting one: it predicts when a commercial relationship will turn adversarial (Domain 10, KA 10.4; Domain 11's negotiation).

7.4.4 Cash flow versus profit

A project can be profitable and still fail for lack of cash — the same truth PFL-AI establishes for financings (its Domain 1). For a delivery leader the mechanism is payment terms: cost is incurred as work happens; cash arrives when invoices are approved and paid. On Auriga, with `AC` = USD 2,120,000 at week 13 and 60-day terms, roughly **USD 742,000** of incurred cost (about 35 %) is still unpaid — an exposure carried by whoever is funding the work. The leader's obligations are practical: know the terms in both directions (client and subcontractors), front-load nothing you cannot fund, and never let a retention or milestone-payment structure be agreed without someone computing its cash profile. Where the project sits inside a financed asset, this is precisely the CFADS conversation PFL-AI Domain 10 has with lenders.

AI in this KA

Contract analytics — extracting terms, comparing models, flagging inconsistent clauses — is real and useful AI assistance, and it is also decision support rather than legal or commercial advice. Two boundaries: an AI-produced reading of a clause is verified against the clause itself before anyone relies on it (the document-against-summary check), and commercial and legal positions go to qualified counsel (Domain 10). Fee arithmetic and PTA computations are deterministic — they get the same golden-answer treatment as the earned-value set.

KEY TERMS — KA 7.4

Term	Meaning
Blended rate	Weighted average cost per hour of a mixed resource pool.
FFP / FPIF / CPFF / CPIF / T&M	Contract models ordered by who carries cost risk.
Share ratio	The agreed buyer/seller split of over- and under-run.
Ceiling price	The cap on the buyer's total exposure under an incentive contract.
Point of total assumption (PTA)	The cost above which the seller bears 100 % of further overrun.
Retention	A withheld percentage of payment, released on completion criteria.

SAMPLE MCQS — KA 7.4

MCQ 7.4-A `[7.4.3 · Application]` Target cost 2,000,000; target fee 150,000; share 70/30; actual cost 2,300,000. The seller's fee is: - A. USD 150,000 - B. USD 60,000 - C. USD 90,000 - D. USD 45,000

Rationale: The seller absorbs 30 % of the 300,000 overrun: `150,000 − 90,000 = 60,000`. A ignores the incentive; C states the fee reduction rather than the fee; D applies the buyer's share to the fee.

MCQ 7.4-B `[7.4.3 · Analysis]` Target cost 2,000,000, target fee 150,000, ceiling 2,450,000, buyer share 70 %. The PTA is 2,428,571, and its delivery significance is that above it: - A. the contract becomes void - B. the buyer absorbs all further cost - C. the seller bears 100 % of further cost, so the incentive inverts and cost growth becomes a scope-change argument - D. the fee becomes negative but risk-sharing continues unchanged

Rationale: Beyond the PTA the ceiling binds the buyer, so every further dollar is the seller's — which predictably redirects the seller's effort from efficiency to entitlement. B reverses the exposure; A is fiction; D misses that sharing has stopped.

MCQ 7.4-C `[7.4.2 · Analysis]` A leader lets an FFP contract for scope that is only 30 % defined. The most likely outcome is: - A. cost risk is genuinely eliminated - B. a priced-in risk premium plus a claims-and-variations exposure as the scope is defined - C. the seller absorbs all scope growth at no cost to the buyer - D. the contract converts automatically to cost-plus

Rationale: Fixed price transfers risk at a price and only for the scope actually specified; undefined scope returns as variations. A and C mistake the contractual form for the underlying uncertainty; D invents a mechanism.

MCQ 7.4-D `[7.4.1 · Application]` A pool of 40 at USD 95/h, 25 at USD 140/h and 15 at USD 210/h has a blended rate of: - A. USD 148.33 - B. USD 130.63 - C. USD 115.00 - D. USD 140.00

Rationale: `10,450/80 = 130.63`. A averages the three rates unweighted; C weights toward the cheapest grade only; D takes the middle rate as representative.

SELF-CHECK — KA 7.4

1. State the one question every contract model answers. — Who carries cost risk.

2. Why is the *PTA* a delivery concern, not just a commercial one? — Above it the seller bears all further cost, so behaviour shifts from efficiency to entitlement.
3. How can a profitable project run out of cash? — Cost is incurred as work happens; cash arrives on payment terms — the gap must be funded.

Advanced topics — Domain 7

7.A.1 Earned schedule — closing `SPI`'s late-project blind spot

`SPI` converges on 1.00 as a project finishes, whatever its lateness, because `EV` and `PV` both approach `BAC`. **Earned schedule** restates progress in time: `ES` is the date at which the value now earned was planned to have been earned, and `SPI(t) = ES / AT`. Auriga's week-13 status, with value earned that the baseline expected by week 12, gives `SPI(t) = 12/13 = 0.92` — and unlike `SPI` it stays meaningful to the last week. This is the bridge flagged in Domain 6 (KA 6.4.3), now with its cost-side companions.

7.A.2 EVM's limitations, stated plainly

Earned value measures conformance to plan, not value to the customer: a project can score 1.00 on both indices while building the wrong thing (Domain 5's acceptance criteria are the defence). It says nothing about quality (Domain 9). It is only as honest as its earning rules and its accruals. And it is blind to risk that has not yet materialised — Domain 8's contingency analysis, not `CPI`, tells you whether what remains is adequately funded. EVM is a measurement system, not a management philosophy; leaders who treat the indices as targets get optimised indices.

7.A.3 The reviewer's cost eye

Invariants a reviewer runs before trusting a cost report: `EV/BAC` equals the claimed percent complete; `CV = EV − AC` and `SV = EV − PV` reconcile to the stated indices; `EAC` = `AC` + a stated `ETC`, with the assumption named; `TCPI` to `BAC/CPI` equals `CPI`; sum of control-account budgets + contingency equals `BAC`; management reserve sits outside that total; no completed package's budget has moved since last period; accruals are present in the current period; and level-of-effort share is disclosed per control account. Any violation is a defect somewhere — find it before the board builds on it.

Industry variations — Domain 7

- **Construction and EPC.** Full EVM against a resource-loaded baseline; measured-work valuations and retention dominate cash; FIDIC-style variation and claims machinery makes the `PTA` conversation routine.
- **Government and defence programmes.** Formal EVMS compliance with surveillance, strict LOE caps and control-account discipline; forecasting is auditable rather than discretionary.
- **Technology and product delivery.** Cost is overwhelmingly people, so blended rates and capacity, not materials, drive the number; where cadence replaces a fixed baseline, throughput and cost-per-increment stand in for `CPI` (Domain 13), and hybrid programmes report both.

- **Energy and resources.** Long-lead equipment and commodity exposure put escalation and currency at the centre of the estimate (PFL-AI Domain 3's machinery); contingency is sized quantitatively as a matter of course.
- **Public services and transformation.** Benefits, not cost variance, are the accountability currency (Domain 16); the leader's task is keeping cost reporting honest while the value case is what gets debated.

Case study — Domain 7: the forecast the board actually needed (utilities)

Situation. Auriga's week-13 review. The cost report shows CPI 0.91, SPI 0.92, CV (200,000), SV (160,000). The programme director's paper proposes reporting EAC **USD 4,200,000** — method (a) — on the grounds that the overrun was the contaminated-ground remediation, a discrete event now closed and remediated (Domain 6's case study).

The challenge. The assurance reviewer asks three questions. Is the cause genuinely closed? Yes for the remediation — but the recovery bought a second civil crew and a fast-track of installation, both of which continue for six more weeks, and neither is in method (a)'s "remaining work at budgeted rate" assumption. What does recovery to *BAC* require? TCPI 1.11 against a demonstrated 0.91. What is the mechanism? There isn't one — the recovery spend increases cost to protect the date.

The outcome. The board is given a **range with named assumptions**: 4,200,000 if the closed event is the whole story; 4,416,667 if current efficiency persists; and a bottom-up (method d) re-estimate of the remaining 52 %, commissioned that week, as the number they will actually manage to. Contingency draw is authorised against the identified risk; management reserve is not touched. The minute records the TCPI gap and the explicit finding that **no recovery-to-budget plan exists** — so nobody later claims one was promised.

What the domain teaches here. A single-number forecast hides the only thing that matters: the assumption. TCPI converts optimism into an arithmetic claim someone has to defend, and the honest report is the one that survives the question "what would have to be true?"

Case study B — Domain 7: past the point of total assumption (technology)

Situation. A systems-integration subcontract ran on an incentive structure: target cost USD 2,000,000, target fee 150,000, 70/30 share, ceiling 2,450,000 — so PTA USD 2,428,571. By month eight the supplier's internal cost was tracking to 2,600,000, well past the PTA.

What happened. The supplier's behaviour changed abruptly and, in hindsight, rationally: fresh change requests on work previously treated as in-scope, slower responses on defect fixes that earned nothing, and a claim that the buyer's late environment provision had caused the growth. The buyer's team read it as bad faith. It was arithmetic: past the PTA every additional engineer-hour came out of the supplier's own margin, and the only route back to profit was re-characterising cost as buyer-caused scope.

The outcome. The parties re-set commercially — a re-baselined target cost recognising the genuine environment delay (documented, and partly the buyer's), a revised ceiling, and a time-and-materials envelope for the disputed remainder. Delivery recovered within two months of the reset. The retrospective's finding: the buyer had never computed the PTA, so a foreseeable inflection was experienced as a betrayal.

What the domain teaches here. Commercial structures create behaviour. Computing the PTA at signature — and watching the supplier's cost trend against it — turns an adversarial surprise into a managed conversation held early, while both parties still have options (Domain 10's supplier governance; Domain 11's negotiation).

Executive perspective — Domain 7

What a project leader cannot delegate in this domain:

- **The forecast's assumption.** Analysts compute EAC; the leader owns which method and why, in one sentence a board can challenge. Reporting a number without its assumption is the domain's cardinal sin.
- **The TCPI reality test.** Before endorsing any recovery-to-budget plan: what index does the remaining work require, what is being demonstrated, and what specific mechanism closes the gap?
- **Reserve discipline.** Contingency and management reserve kept distinct, spent under stated authority, with consumption trended against progress — the erosion in MCQ 7.1-B is a leadership signal, not a bookkeeping detail.
- **Measurement integrity.** Earning rules fixed in advance, accruals present, level-of-effort share disclosed, no retrospective budget edits. Cost systems fail morally before they fail arithmetically — the same sentence as Domain 6, and the same reason.
- **The commercial shape.** Which model, why, and where the PTA sits. A leader who cannot say who carries cost risk on their largest package is not yet in control of it.

Calculation exercises — Domain 7

Exercise 7.1 BAC 4,000,000; PV 2,080,000; EV 1,920,000; AC 2,120,000. Compute CV, SV, CPI, SPI, percent complete and percent spent. Solution. CV (200,000); SV (160,000); CPI 0.91; SPI 0.92; complete `1,920,000/4,000,000 =` **48.0 %**; spent `2,120,000/4,000,000 =` **53.0 %**. Common error: computing percent complete from AC/BAC — that is percent spent, and reporting it as progress overstates the project.

Exercise 7.2 Same data. Compute EAC by methods (a), (b) and (c), and the corresponding VAC. Solution. (a) `2,120,000 + 2,080,000 =` **4,200,000**, VAC **(200,000)**. (b) `4,000,000/0.905660 =` **4,416,667**, VAC **(416,667)**. (c) `2,120,000 + 2,080,000/(0.905660 × 0.923077) =` **4,608,056**, VAC **(608,056)**. Common error: rounding CPI to 0.91 before dividing — `4,000,000/0.91 = 4,395,604`, USD 21,063 adrift. Indices are display; arithmetic is full precision.

Exercise 7.3 Same data. Compute TCPI to BAC and to the method-(b) EAC, and interpret. Solution. To BAC: `2,080,000/1,880,000 =` **1.11**. To (b): `2,080,000/(4,416,667 − 2,120,000) = 2,080,000/2,296,667 =` **0.91**. The second equals the current CPI — the identity of 7.3.4: BAC/CPI forecasts are "nothing changes". Common error: using PV rather than AC in the denominator (giving 1.08) and concluding recovery is nearly free.

> **Exercise 7.4** Incentive subcontract: target cost 2,400,000; target fee 180,000; share 80/20; ceiling 2,900,000. The seller finishes at 2,650,000. Find the fee, what the buyer pays, and the `PTA`. Solution. Overrun 250,000; fee `180,000 − 250,000 × 0.20 =` **130,000**; buyer pays `2,650,000 + 130,000 =` **2,780,000** (below the 2,900,000 ceiling). Target price 2,580,000; `PTA = 2,400,000 + (2,900,000 − 2,580,000)/0.80 = 2,400,000 + 400,000 =` **2,800,000**. Common error: applying the buyer's 80 % share to the fee reduction.

> **Exercise 7.5** A pool of 30 at USD 105/h, 20 at USD 150/h and 10 at USD 220/h. Compute the blended rate, then the cost of adding four weeks of a 5-person crew at 40 h/week. Solution. `(30×105 + 20×150 + 10×220)/60 = (3,150 + 3,000 + 2,200)/60 = 8,350/60 =` **USD 139.17/h**. Crew cost `5 × 40 × 4 × 139.17 =` **USD 111,333**. Common error: using the blended rate for a crew drawn entirely from one grade — the blend applies to a representative mix, not to any arbitrary subset.

Practitioner's toolkit — Domain 7

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 7.T.1 — Cost-report integrity checklist (run before publishing the month)

- [ ] Earning rules fixed in advance per package and unchanged this period; any change disclosed.
- [ ] `EV` claims sampled against physical evidence, not accepted on assertion.
- [ ] Accruals recognised for work received but not invoiced; open commitments cleansed.
- [ ] `PV` read from the current controlled baseline, phased over the approved schedule.
- [ ] No retro-fitted budgets; transfers between control accounts logged.
- [ ] `EV/BAC` reconciles to the reported percent complete; indices recompute from the four numbers.
- [ ] Every `EAC` states its method **and its assumption**; `TCPI` to `BAC` reported beside it.
- [ ] Contingency and management reserve shown separately, with consumption vs progress.
- [ ] Level-of-effort share disclosed per control account.
- [ ] AI-produced forecasts marked, calibration record attached, human owner named.

Toolkit 7.T.2 — Estimate basis sheet (one per estimate)

Method (analogous/parametric/bottom-up) · definition maturity and accuracy class · range (−x %/+y %) · rate sources with dates and escalation basis · quantities and their source · exclusions stated explicitly · risks feeding contingency (link to the register) · reviewer and date. An estimate whose basis sheet cannot be produced is not releasable.

Toolkit 7.T.3 — Commercial one-pager (per major package)

Model (FFP/FPIF/CPFF/CPIF/T&M) and why · who carries cost risk · target cost, fee, share ratio, ceiling · `PTA` **and current cost trend against it** · payment terms both directions and the cash profile · retention and release criteria · variation and claims route · escalation contacts. Reviewed at every commercial checkpoint, not filed at signature.

Exam preparation — Domain 7

The calculation traps. Percent spent reported as percent complete (Exercise 7.1) · rounding `CPI` before dividing (Exercise 7.2) · `PV` instead of `AC` in the `TCPI` denominator (Exercise 7.3) · applying the wrong party's share to an incentive fee (Exercise 7.4) · `EV` valued at actual cost rather than budget · reading a level-of-effort-heavy `SPI` as schedule performance (MCQ 7.3-D) · quoting `SPI` late in a project where it must converge on 1.00 (7.A.1) · confusing commitments with actuals · treating contingency and management reserve as one pot.

Reflection questions. 1. Your cost report shows one `EAC`. What must accompany it before you will sign it? (The method and the assumption about remaining work; *`TCPI`* to *`BAC`* beside it.) 2. On your largest package: who carries cost risk, where is the `PTA`, and how close is the supplier's cost trend to it? (7.4.2–7.4.3; toolkit 7.T.3.) 3. Which invariant in 7.A.3 would have caught the last cost surprise you experienced — and why wasn't it running?

Domain 7 summary

Cost leadership begins with an honest number: a method matched to definition maturity, a range and class always stated, three-point thinking where tails are real, and a budget built through control accounts into a time-phased baseline with contingency inside it and management reserve outside. Measurement then has to be earned — `AC` with accruals, `EV` at budget under earning rules fixed in advance — because everything downstream is arithmetic on those numbers: `CV`, `SV`, `CPI`, `SPI` at Auriga's week 13 (0.91 and 0.92; 48 % complete against 53 % spent), the `EAC` family spanning USD 408,056 on identical data because they encode different assumptions, and `VAC` and `TCPI` turning recovery talk into a defensible index (1.11 required against 0.91 demonstrated). The commercial half is the same discipline pointed outward: blended rates make options commensurable, contract models allocate cost risk at a price, and the point of total assumption predicts when a supplier's incentives invert. Throughout, the leadership rule holds — the number is only as good as the assumption named beside it, and machine-produced forecasts reach a board only with a calibration record and a human owner. Domain 8 quantifies the risk this domain reserves for; Domain 10 takes the commercial relationships into their own depth.

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